



Enhancing Decision Support in Urban Parking through Real-Time ML Pipelines and Interactive Dashboards: A Case Study from Bamberg

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Abstract

Urban parking congestion is a persistent challenge in modern cities, often leading to increased traffic, longer travel times, and higher levels of air pollution. A significant portion of urban traffic is caused by drivers searching for available parking spaces, highlighting the need for predictive systems that can provide reliable information about future parking availability. This thesis investigates how machine learning techniques can be used to forecast parking availability and support data-driven decision making in urban mobility systems.

To address this problem, several forecasting models were evaluated for multi-step time-series prediction of parking availability using sensor data from two parking facilities. The study compares classical machine learning methods and deep learning approaches and introduces a horizon-aware hybrid forecasting model designed to combine the strengths of different algorithms. In addition, a feature importance analysis was conducted to identify the variables that most strongly influence parking availability prediction.

The results show that forecasting accuracy varies across prediction horizons and that the proposed hybrid model consistently outperforms individual baseline models in terms of prediction error. The analysis further reveals that temporal variables and recent occupancy statistics are the most influential predictors, while weather variables play a comparatively smaller role. Finally, the forecasting framework was integrated into an automated data pipeline and deployed as a real-time predictive dashboard, demonstrating the practical feasibility of combining machine learning forecasting with operational decision-support systems for urban parking management.

Contents

1	Introduction	1
1.1	Motivation and Problem Statement	1
1.2	Research Gap	2
1.3	Research Objectives	3
1.4	Research Questions	4
1.5	Research Scope	4
1.6	Thesis Structure	5
2	State of Research	7
2.1	Data Inputs and Feature Engineering	7
2.2	Target Formulation and Definition of Parking Availability	10
2.3	Applied Machine Learning and Deep Learning Models	12
2.3.1	Classical Machine Learning Approaches	12
2.3.2	Deep Learning and Hybrid Neural Approaches	13
2.4	Research Gaps and Opportunities	14
3	Method	17
3.1	Dataset and Feature Engineering	17
3.1.1	Parking Availability Data	18
3.1.2	Weather Data	19
3.1.3	Exploratory Data Analysis	20
3.1.4	Data Preprocessing and Feature Construction	23
3.2	Model Development and Evaluation	27
3.2.1	Classical Machine Learning Models	28
3.2.2	Deep Learning Models	30
3.2.3	Horizon-Aware Hybrid Model	32
3.2.4	Feature Importance Analysis	36
3.2.5	Implementation Environment	37
3.2.6	Evaluation Metrics	38
3.3	Deployment and System Architecture	39
3.3.1	Data Collection Component	40
3.3.2	Data Transformation Component	41
3.3.3	Prediction Component	42
3.3.4	Interactive Dashboard Component	42

CONTENTS

4	Results and Evaluation	45
4.1	Overall Baseline Models Performance	46
4.2	Performance Across Forecast Horizons	47
4.3	Hybrid Model vs Baselines	49
4.3.1	Feature Importance Analysis	52
4.4	Deployment Results: Interactive Forecast Dashboard	54
4.4.1	Predicted Parking Availability Visualization	55
4.4.2	Historical and Predicted Parking Availability Trends . . .	56
4.4.3	24-Hour Temperature Forecast Visualization	57
5	Discussion and Conclusion	61
5.1	Discussion	61
5.1.1	Interpretation of the Forecasting Results	61
5.1.2	Answering the Research Questions	63
5.1.3	Theoretical Implications	65
5.1.4	Practical Implications	65
5.2	Conclusion and Future Work	67
	Bibliography	73
	A Qualitative Prediction Comparison	75
	Declaration of Authorship	77

List of Tables

3.1	Predictive models and key hyperparameters.	28
4.1	Baseline model performance for Lot 38 (24-hour forecast).	46
4.2	Baseline model performance for Lot 634 (24-hour forecast).	47
4.3	Overall performance of all evaluated models for Lot 38.	53
4.4	Overall performance of all evaluated models for Lot 634.	55

List of Figures

3.1	Overview of the proposed real-time deployment architecture. . .	18
3.2	Raw parking event data (PostgreSQL excerpt).	19
3.3	Hourly weather data (PostgreSQL excerpt).	20
3.4	Mean hourly parking availability for Lot 38 and Lot 634 with interquartile range.	21
3.5	Average parking availability by weekday for Lot 38 and Lot 634 (bars: mean, error bars: variability).	22
3.6	Average parking availability by month for Lot 38 and Lot 634. .	23
3.7	Hourly parking availability (Lot 634) stored in PostgreSQL. . . .	25
3.8	Integrated training dataset (hourly).	25
3.9	Multi-output forecasting: 48 past time steps mapped to 24 future predictions.	29
3.10	Deep learning forecasting architectures: sequence models (RNN/CNN) and encoder–decoder for direct multi-step prediction.	30
3.11	Validation RMSE vs. α for GWB (Lot 38).	35
3.12	Validation RMSE vs. α for GWB (Lot 634).	35
3.13	Horizon-specific optimal weights α_h for the LWB model (Lot 38). .	36
3.14	Horizon-specific optimal weights α_h for the LWB model (Lot 634). .	37
3.15	Containerized cloud deployment architecture.	40
4.1	RMSE by forecast horizon for the top four baseline models on Lot 38.	49
4.2	Horizon-wise best model for Lot 38 (lowest RMSE).	50
4.3	RMSE by forecast horizon for the top four baseline models on Lot 634.	51
4.4	Horizon-wise best model for Lot 634 (lowest RMSE).	52
4.5	Horizon-wise best model for Lot 38.	54
4.6	Horizon-wise best model for Lot 634.	56
4.7	Top 15 important features for Lot 38 (Random Forest).	57
4.8	Top 15 important features for Lot 634 (Random Forest).	58
4.9	Web dashboard for real-time parking availability and 24-hour forecasts.	59
4.10	Predicted parking availability for Lots 38 and 634.	59
4.11	Historical (48h) and predicted (24h) parking availability for Lots 38 and 634.	59
4.12	24-hour temperature forecast for Bamberg.	60
A.1	Example predictions vs. actual values for top models (Lot 38). .	75

LIST OF FIGURES

A.2 Example predictions vs. actual values for top models (Lot 634). . .	76
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Chapter 1

Introduction

1.1 Motivation and Problem Statement

In many urban areas, population growth and increasing car ownership have significantly intensified the demand for parking. However, parking infrastructure has not expanded at the same rate, resulting in a structural imbalance between demand and available capacity. As a consequence, finding a free parking space has become increasingly difficult, particularly during peak hours in dense city centers. In such situations, drivers are often required to circulate across nearby streets or multiple parking facilities before successfully parking. This behavior transforms parking from a simple convenience issue into a broader urban mobility challenge Shoup (2006); Arnott and Inci (2006).

A substantial body of research highlights that a significant portion of urban traffic congestion is not generated by vehicles in transit, but by drivers searching for parking spaces. This phenomenon, commonly referred to as cruising for parking, contributes directly to increased travel time, fuel consumption, and environmental emissions Ma et al. (2024); Polycarpou et al. (2013).

Empirical studies further emphasize the scale of this issue. For instance, studies report that up to 30% of inner-city traffic can be attributed to parking search behavior, while drivers collectively spend billions of hours annually searching for parking spaces, resulting in significant economic and environmental costs INRIX (2017). In Germany specifically, parking search has been associated with substantial fuel waste and economic losses, underlining its relevance as a national urban mobility concern.

At the core of this challenge lies the concept of parking availability, typically defined as the number of free parking spaces within a facility or area at a given time. This metric reflects the dynamic interaction between fluctuating demand patterns and limited infrastructure capacity.

Traditional parking information systems primarily provide real-time occupancy data, often derived from sensor-based monitoring systems, allowing users to observe current parking conditions Caicedo (2010). While such systems improve situational awareness, they remain inherently reactive.

However, parking decisions are fundamentally time-dependent. Drivers frequently plan their trips, destinations, and parking choices in advance, especially in commuting or event-driven scenarios. In this context, knowledge of current

availability alone is insufficient, as it does not account for future demand fluctuations or expected occupancy at the time of arrival Bischoff and Nagel (2017); Ma et al. (2024).

Recent advances in sensing technologies and data infrastructure have enabled the continuous collection of high-resolution parking data, creating opportunities to model temporal patterns and develop predictive solutions Channamallu et al. (2023a). From a practical perspective, short-term forecasts of parking availability, particularly over a 24-hour horizon, provide significantly greater value than real-time information alone. Such forecasts allow users to make informed, proactive decisions rather than reactive adjustments upon arrival Ma et al. (2024); Bischoff and Nagel (2017).

This shift toward temporally informed parking guidance aligns with broader developments in smart city initiatives, where data-driven decision support systems are increasingly deployed to improve efficiency, sustainability, and user experience. As such, short-term parking availability prediction represents not only an active research topic within intelligent transportation systems, but also a practically relevant problem with direct implications for urban mobility management and real-world decision support

1.2 Research Gap

Despite extensive research on sensor-based parking availability prediction, two fundamental limitations remain insufficiently addressed in the existing literature. First, the majority of studies implicitly assume that a single predictive model can adequately serve the entire multi-step forecasting horizon, even though empirical evaluations consistently indicate that model performance varies substantially across lead times Zhao et al. (2020); Awan et al. (2020); Lyu et al. (2024). Simpler statistical or regression-based models often achieve lower errors at very short horizons, where demand dynamics are dominated by strong local trends and regular temporal patterns, whereas more complex machine learning or deep learning models may perform better at medium or longer horizons by capturing nonlinear dependencies and broader temporal structures Ma et al. (2024); Channamallu et al. (2023a). However, most existing approaches overlook this horizon-dependent behavior and deploy a single model to generate predictions for all forecast steps, effectively averaging performance across horizons and masking systematic strengths and weaknesses. As a result, the predictive potential of complementary models remains underexploited, and forecast accuracy is not optimized for the specific decision times that are most relevant to drivers.

Second, while a growing body of work reports promising predictive accuracy under offline evaluation settings, far fewer studies address how these models can be operationalized as part of an end-to-end parking decision support system Zhao et al. (2020); Awan et al. (2020). In particular, machine learning models are often treated as isolated analytical components rather than as integral elements of a real-time pipeline that continuously ingests sensor data, performs inference, and communicates results to end users. Existing implementations are typically limited to backend services or static visualizations, and only a small number of contributions demonstrate how predictive outputs are translated into interactive, user-facing tools such as dashboards that explicitly support parking-related decisions Ma et al. (2024); Bischoff and Nagel (2017);

Channamallu et al. (2023a). Consequently, parking prediction is rarely framed as a decision-support problem from the driver’s perspective, where forecasts should directly inform arrival-time planning and facility selection. This separation between predictive modeling and practical system deployment limits the real-world impact of parking availability forecasting, as users are rarely exposed to predictive insights in a form that is actionable and temporally meaningful.

Taken together, these limitations define the core research problem addressed in this thesis. There is a lack of horizon-aware forecasting approaches that explicitly account for varying model performance across short-term prediction horizons, and an equally important lack of deployable, dashboard-based systems that integrate machine learning forecasts into real-time parking decision support. Addressing this problem is essential for moving beyond purely methodological comparisons and toward operational solutions that treat machine learning as a functional component of urban parking information systems. This thesis responds to these challenges by developing a horizon-aware hybrid forecasting approach and embedding it within a real-time data pipeline and interactive dashboard, thereby bridging the gap between predictive accuracy, system deployment, and user-centered decision support.

1.3 Research Objectives

The overarching aim of this thesis is to enhance decision support in urban parking systems by developing and deploying a real-time machine learning–based forecasting framework that is both technically robust and practically usable. The focus lies on predicting parking availability over a 24-hour horizon and integrating these predictions into an operational, dashboard-based decision-support system for drivers. Using sensor-based occupancy data from two off-street parking facilities in Bamberg, Germany, the thesis moves beyond purely analytical evaluation and treats parking prediction as a system-level problem, where forecasting models, data pipelines, and user-facing interfaces jointly contribute to informed arrival-time planning. In this context, machine learning is not considered an isolated modeling exercise, but a functional component of a real-time parking information system designed to support urban mobility decisions.

To achieve this aim, the thesis follows a set of closely connected objectives that span data analysis, modeling, system design, and deployment. First, sensor-based parking occupancy data are analyzed to capture temporal demand patterns relevant for short-term forecasting. Second, a range of baseline forecasting models, including classical machine learning and deep learning approaches, are implemented and evaluated under a consistent experimental protocol to assess their performance across multiple forecast horizons. Building on these results, the thesis develops a horizon-aware hybrid forecasting strategy that explicitly accounts for horizon-dependent model performance through global and local weighted blending. In order to better understand the factors influencing parking availability prediction, a feature importance analysis is conducted to identify the most influential variables and to provide interpretability regarding the contribution of temporal, occupancy-based, and contextual features. Beyond model development and interpretation, the selected forecasting approach is operationalized within a real-time data engineering pipeline that supports continuous data ingestion and inference. Finally, the predictive outputs are

translated into an interactive dashboard that visualizes expected parking availability over the next 24 hours, enabling user-oriented decision support rather than purely offline performance assessment.

1.4 Research Questions

These objectives give rise to a set of research questions that guide the methodological design of the thesis. The research questions focus on both the predictive performance of different forecasting approaches and the practical integration of such models into a real-world decision-support system. Specifically, the thesis addresses the following research questions:

1. **RQ1: Which variables have the greatest impact on parking availability prediction?**
2. **RQ2: How does forecasting performance vary across different prediction horizons?**
3. **RQ3: Can a horizon-aware hybrid forecasting model improve predictive accuracy compared to single-model approaches?**
4. **RQ4: How can a parking availability forecasting framework be integrated into a real-time data pipeline that ensures continuous operation and reproducibility?**
5. **RQ5: How can a predictive parking dashboard support decision-making for drivers, city planners, and parking operators?**

Together, these research questions ensure that the thesis addresses both methodological and practical dimensions of parking availability forecasting, linking predictive accuracy with model interpretability, real-world deployment, and decision-support capabilities.

1.5 Research Scope

The scope of this thesis is deliberately focused on short-term parking availability forecasting and decision support in an urban off-street parking context. Empirical analysis and system implementation are based on sensor-derived occupancy data from two off-street parking facilities in the city of Bamberg, Germany. These facilities were selected due to data availability, stable sensor coverage, and their relevance for commuter-oriented parking demand. The geographical scope is therefore limited to a single mid-sized European city, allowing for an in-depth, context-specific investigation while maintaining practical relevance for comparable urban environments with similar parking infrastructure.

From a temporal perspective, the thesis concentrates on short-term forecasting within a 24-hour horizon. This horizon was chosen because it aligns with typical driver decision-making processes related to arrival-time planning and daily travel routines. Longer-term forecasting horizons, such as weekly or monthly predictions, are outside the scope of this work, as they are more strongly influenced by exogenous factors such as seasonal trends, policy changes, or long-term

mobility shifts. The focus on short-term horizons enables a detailed examination of horizon-dependent model performance and supports the integration of forecasts into real-time decision-support systems.

In terms of methodological scope, the thesis evaluates a selected set of statistical, machine learning, and deep learning models that are commonly used in parking occupancy prediction and short-term time series forecasting. Model selection is guided by prior literature and practical deployability rather than exhaustive algorithmic coverage. While the study introduces a horizon-aware hybrid forecasting approach based on weighted blending of top-performing models, it does not explore more complex ensemble techniques such as stacking with meta-learners or probabilistic forecasting methods. The emphasis remains on interpretability, robustness, and suitability for real-time deployment within a constrained system architecture.

1.6 Thesis Structure

This thesis is structured into five main chapters. Chapter 2 reviews the state of research, covering data inputs and feature engineering, forecasting problem formulation, applied machine learning and deep learning models, and identified research gaps. Chapter 3 presents the methodological framework, including dataset construction, feature engineering, model development and evaluation, and the design of the real-time deployment architecture with its core system components. Chapter 4 reports the experimental results and evaluates model performance, including baseline comparisons, horizon-wise analysis, hybrid model evaluation, and deployment outcomes such as the interactive dashboard. Finally, Chapter 5 discusses the findings, interprets the results in relation to the research questions, outlines theoretical and practical implications, and concludes with directions for future work.

Chapter 2

State of Research

This section reviews existing research on parking availability forecasting and structures the literature around four key dimensions. First, it examines the types of data inputs and feature engineering strategies used to capture parking dynamics. Second, it discusses how the forecasting task is formulated and how parking availability is defined across studies. Third, it reviews the range of applied machine learning and deep learning models. Finally, it identifies key research gaps and opportunities, which motivate the horizon-aware approach proposed in this thesis.

2.1 Data Inputs and Feature Engineering

Across recent studies on parking availability forecasting, the nature and granularity of available data have fundamentally shaped methodological choices, feature engineering strategies, and achievable predictive performance. Data sources range from relatively simple parking-session logs to dense city-scale sensor networks with minute-level reporting frequencies, and this heterogeneity directly influences how availability dynamics are modeled. Differences in temporal resolution, spatial coverage, and the availability of contextual variables determine both the complexity of forecasting models and the type of patterns that can be captured.

Early work in parking prediction frequently relied on structured parking-event records containing identifiers and start/end timestamps, such as parking spot ID, date, and minute-level arrival and departure information. These transactional datasets allow parking availability to be reconstructed indirectly by aggregating individual parking sessions over time. For example, Awan et al. analyze parking event records to derive temporal usage patterns and evaluate forecasting methods based on reconstructed availability signals Awan et al. (2020). In such datasets, predictive modeling typically focuses on capturing regular temporal structures such as daily peak periods, weekday-weekend differences, and seasonal variations. Because availability is inferred from transactional data rather than directly observed, feature engineering often emphasizes temporal indicators including hour of day, day of week, and calendar effects.

A second and increasingly common category of datasets relies on direct availability time series derived from dedicated parking sensors. These datasets typ-

ically consist of binary or aggregated availability states collected at regular intervals across individual parking spaces or entire parking facilities. Compared to transactional logs, sensor-based datasets provide continuous observations of parking conditions and allow models to capture short-term dynamics more effectively. For instance, Zhao et al. analyze availability signals collected from multiple parking facilities and demonstrate that historical availability alone can serve as a strong predictive signal Zhao et al. (2020). In such contexts, autoregressive patterns, daily periodicity, and weekly cycles frequently dominate the predictive signal. As a result, feature engineering strategies often include lagged availability values, rolling averages, and calendar-based indicators to capture short-term demand fluctuations.

High-resolution sensor deployments further extend this paradigm by providing fine-grained temporal observations of availability dynamics. Vlahogianni et al. describe deployments involving hundreds of sensors reporting minute-level parking states, enabling the analysis of rapid changes in parking demand Vlahogianni et al. (2016). Their results indicate that relatively short look-back windows—sometimes as small as five to ten minutes—can already capture meaningful short-term dynamics for real-time forecasting tasks. Such high-frequency datasets are particularly well suited for short-horizon prediction, where recent availability observations provide strong predictive signals and models can respond quickly to sudden demand shifts.

With the advancement of smart city infrastructures, more recent studies increasingly incorporate contextual variables to enrich the predictive signal. These enriched datasets combine parking availability data with external information such as traffic conditions, weather variables, and roadway characteristics. For example, Yang et al. integrate parking-meter transactions with traffic speed and roadway descriptors and report that traffic-related variables often exert stronger predictive influence than meteorological conditions, although weather variables such as temperature and precipitation still contribute measurable improvements Yang et al. (2019). Similarly, Bilotta et al. investigate multi-step forecasting scenarios with prediction horizons extending up to 24 hours and incorporate historical availability data together with traffic indicators and meteorological variables including humidity, speed, and travel time Bilotta et al. (2023). Their findings suggest that while short-term predictions can often rely heavily on autoregressive availability signals, exogenous variables become increasingly important as prediction horizons grow longer.

Beyond individual case studies, several review papers highlight the diversity of data sources used in parking availability prediction research. Channamallu et al., in their systematic review of parking prediction methods, categorize datasets into on-street sensor networks, off-street garage monitoring systems, and transaction-based parking meter records Channamallu et al. (2023b). Their analysis indicates that data availability strongly influences methodological choices. While availability time series remain the most common input representation, there is a growing trend toward integrating heterogeneous data sources such as weather services, traffic sensors, and event information in order to improve model robustness under irregular demand conditions. Related work Channamallu et al. (2024) further emphasizes that the digitalization of parking services and urban mobility infrastructures has enabled real-time data collection at unprecedented scales, thereby facilitating predictive analytics and automated parking guidance systems.

Publicly available datasets have also contributed to methodological standardization in the field. Domenech et al. describe an AI-based parking forecasting system using publicly released CSV datasets containing car park identifiers and availability expressed as a percentage of free spaces Domenech et al. (2025). In these datasets, engineered sliding windows—often spanning a full 24-hour cycle—are constructed to capture periodic demand patterns and support multi-step forecasting tasks. Although such standardized datasets improve reproducibility and comparability across models, they may also abstract away some of the irregularities present in raw sensor data.

Recent research has also expanded the scope of input representation by incorporating spatial relationships between parking facilities. In dense urban environments, parking demand in one location can influence neighboring facilities as drivers search for available spaces. Sasaki et al. propose a graph-to-sequence forecasting model that represents parking facilities as nodes within a spatial graph and models interactions between them using graph neural networks Sasaki et al. (2022). Similarly, studies integrating graph convolutional networks (GCNs) with recurrent architectures demonstrate improved predictive performance in large-scale deployments where spatial correlations between parking facilities are significant.

Another emerging dataset category arises from IoT-based architectures and digital twin frameworks. For example, Piccialli et al. describe a digital twin architecture that integrates parking sensors, contextual data sources, and predictive analytics within a unified system Piccialli et al. (2025). Such systems illustrate how availability monitoring, environmental sensing, and predictive modeling can be embedded directly within operational smart city platforms rather than remaining confined to isolated research environments.

Overall, the literature reveals several dominant paradigms in the design of input data for parking availability forecasting: event-based transactional logs, direct availability time series from sensor deployments, enriched multi-modal datasets combining availability with contextual variables such as traffic and weather information, and spatially structured datasets supporting graph-based modeling approaches. The predictive performance reported in each study is closely linked to these underlying data representations. While richer contextual information can improve forecasting robustness—particularly for longer prediction horizons—many studies demonstrate that high-resolution historical availability data alone can yield strong predictive performance for short-term forecasting tasks when temporal regularity is pronounced.

This diversity in dataset design highlights the importance of clearly specifying input representations and feature engineering strategies when comparing forecasting approaches. Differences in temporal granularity, spatial coverage, and exogenous variable inclusion may account for a substantial portion of the performance variation reported across studies. Consequently, understanding how data sources shape feature engineering and model selection is essential for interpreting empirical findings in parking availability forecasting and for designing deployment-ready systems that balance predictive accuracy with realistic data availability constraints.

2.2 Target Formulation and Definition of Parking Availability

Across the parking availability forecasting literature, the definition of the prediction target varies substantially in both semantic meaning and temporal granularity. These variations strongly influence evaluation metrics, model selection, and what constitutes “good performance.” Consequently, differences in target formulation complicate cross-study comparisons and affect how forecasting systems can be integrated into operational decision-support tools.

One common formulation treats parking availability as a binary classification problem at the individual parking spot level. In this setting, a parking space is labeled as either *free* or *occupied* within a predefined short time window. For example, Awan et al. define availability based on whether the utilization rate within a 10–20 minute window exceeds a specified threshold, enabling short-horizon predictions with similar temporal validity Awan et al. (2020). Such formulations are particularly suitable for micro-level parking guidance systems where drivers require near-immediate information about space availability. Binary targets simplify model training and evaluation by enabling the use of standard classification metrics such as accuracy, precision, recall, and F1-score. However, they also abstract away information about partial capacity utilization and do not directly reflect facility-level availability dynamics. Moreover, short-window classification may not adequately support planning-oriented decisions, where knowing whether a facility will have 5% or 40% available capacity could meaningfully influence driver behavior.

A second widely used formulation models parking availability as a continuous regression problem, where the number or proportion of available spaces is predicted as a real-valued time series. Zhao et al., for example, forecast future parking conditions by modeling availability signals directly within a time-series regression framework Zhao et al. (2020). Similarly, Yang et al. focus on short-term regression horizons, such as predicting parking availability 30 minutes ahead Yang et al. (2019). In such settings, model performance is typically evaluated using error-based metrics including root mean squared error (RMSE), mean absolute error (MAE), or mean absolute percentage error (MAPE). Regression formulations preserve more information than binary classification and are particularly appropriate for off-street parking facilities, where decision-making depends on the expected proportion of available spaces rather than the state of individual parking spots. However, regression models may exhibit increased variance at longer prediction horizons and may require normalization when facilities differ significantly in capacity.

Beyond single-step prediction, a substantial portion of the literature addresses multi-step forecasting problems, where models generate predictions for several future time intervals. Vlahogianni et al. extend prediction horizons from one step up to thirty steps ahead, demonstrating how forecasting performance typically deteriorates as lead time increases Vlahogianni et al. (2016). Multi-step forecasting is generally implemented using either recursive strategies—where each predicted value becomes input for the next step—or direct multi-horizon approaches, where the model outputs the entire prediction sequence simultaneously. Recursive methods are simpler to implement but are prone to error accumulation over longer horizons. In contrast, direct multi-output models can

2.2. TARGET FORMULATION AND DEFINITION OF PARKING AVAILABILITY

capture horizon-specific patterns more effectively but typically require more complex architectures and larger training datasets.

In facility-level and off-street parking contexts, targets are frequently defined as the percentage of available spaces relative to total parking capacity. Domenech et al., for instance, employ availability expressed as a percentage of free spaces, enabling consistent normalization across facilities of different sizes Domenech et al. (2025). This representation facilitates comparison across parking facilities and reduces sensitivity to absolute capacity differences. Alternatively, some studies represent availability as a normalized value between 0 and 1, effectively modeling the proportion of available spaces within a facility Koumetio Tekouabou et al. (2022). Such normalized targets support transferability across locations and can improve numerical stability during model training.

Another formulation models parking availability as a count variable, directly predicting the number of free parking spaces. Bilotta et al. adopt this approach in their deep-learning-based forecasting framework, predicting the number of available spaces at 15-minute intervals across a 24-hour horizon Bilotta et al. (2023). Count-based targets align closely with user-oriented applications, where drivers are primarily interested in how many spaces are likely to be available rather than abstract percentages. However, count prediction introduces heteroscedasticity challenges, as prediction errors may scale with facility capacity.

Recent deep learning approaches increasingly focus on mid-term sequence forecasting, where models predict availability across an entire 24-hour horizon at fixed temporal granularity (for example, 15-minute intervals). In these setups, models are typically trained using historical context windows—often spanning one week of past observations—to generate multi-step availability sequences Bilotta et al. (2023). These architectures treat forecasting as a sequence-to-sequence learning problem and frequently incorporate contextual variables such as weather conditions, traffic patterns, and calendar features. While such approaches align closely with planning-oriented applications, they also introduce additional complexity in model evaluation. Aggregated error metrics over a full prediction horizon may obscure systematic differences between short- and long-term forecast accuracy.

Another important dimension concerns the level of spatial aggregation. Some studies operate at the individual parking space level, predicting binary availability states. Others aggregate data across entire facilities or urban districts. Aggregated availability signals typically exhibit smoother temporal dynamics and stronger periodic patterns compared to individual space-level signals, which may fluctuate rapidly due to vehicle turnover. As a result, regression models often appear more stable at facility level, whereas classification approaches are more commonly applied at individual space granularity.

Temporal resolution further complicates comparisons across studies. Prediction intervals in the literature range from five-minute forecasts to hourly predictions. High-frequency forecasts (e.g., 5–10 minutes) are well suited for real-time parking guidance systems but may exhibit greater noise sensitivity. In contrast, hourly forecasts smooth short-term fluctuations and better capture broader demand patterns, although they may be less responsive to sudden changes in parking demand.

Taken together, the literature reveals that parking availability forecasting targets span a broad spectrum of formulations, including short-window binary classification, continuous single-step regression, multi-step recursive forecast-

ing, normalized percentage prediction, count-based modeling, and full-horizon sequence generation. These design choices fundamentally shape evaluation metrics, reported accuracy levels, and system applicability. Importantly, differences in target formulation often obscure model comparisons, as performance improvements reported under one formulation may not translate directly to another.

This heterogeneity highlights the importance of clearly specifying prediction targets and forecast horizons when designing parking forecasting systems. It also underscores the need to evaluate model performance across multiple prediction steps rather than relying solely on aggregated error metrics. In particular, when multi-step forecasting is used for decision-support applications—such as predicting parking availability over a 24-hour horizon—performance variability across lead times becomes a central concern. Recognizing this variability motivates more nuanced evaluation strategies and supports the development of horizon-aware modeling approaches that explicitly account for changing prediction uncertainty over time.

2.3 Applied Machine Learning and Deep Learning Models

Prior work on parking availability forecasting spans a broad spectrum of predictive model families, reflecting the diversity of available datasets, forecasting targets, and deployment objectives. Broadly, existing approaches can be grouped into two methodological categories: (i) classical machine learning models that rely on structured feature representations, and (ii) deep learning architectures designed to capture temporal and spatiotemporal dependencies in availability dynamics. While model complexity has increased in recent years, empirical evidence across multiple studies suggests that forecasting performance remains strongly dependent on dataset characteristics, feature design, and prediction horizon rather than model architecture alone Awan et al. (2020); Vlahogianni et al. (2016); Channamallu et al. (2023c); Koumetio Tekouabou et al. (2022); Bilotta et al. (2023).

2.3.1 Classical Machine Learning Approaches

Classical machine learning (ML) approaches continue to play a significant role in parking availability prediction research. These models typically operate on tabular feature representations derived from historical parking data. Common input features include lagged availability values, temporal indicators such as hour of day and day of week, and contextual variables including weather or traffic information. Frequently evaluated models include k-nearest neighbors (kNN), decision trees, random forests, gradient boosting methods, multilayer perceptrons (MLP), and support vector regression (SVR).

Several empirical studies demonstrate the effectiveness of tree-based ensemble methods in this context. Awan et al. conduct a comparative analysis of multiple machine learning classifiers and regressors and report that decision trees and ensemble variants often achieve strong predictive performance across different experimental settings Awan et al. (2020). Their results highlight that relatively simple learners can outperform more complex alternatives when the

2.3. APPLIED MACHINE LEARNING AND DEEP LEARNING MODELS

feature representation effectively captures dominant temporal patterns in parking demand. Similar observations are reported by Sneha Channamallu et al., whose benchmarking study shows that ensemble-based approaches frequently emerge as top-performing models when applied to structured parking datasets Sneha Channamallu et al. (2024). Their analysis emphasizes the ability of ensemble models to capture nonlinear interactions between temporal and contextual features while remaining robust to noise in real-world sensor data.

Kernel-based learning methods have also demonstrated strong performance in parking forecasting tasks. Koumetio Tekouabou et al. evaluate support vector regression alongside several ensemble models and report competitive performance when the dataset includes both autoregressive availability features and contextual inputs Koumetio Tekouabou et al. (2022). Their findings indicate that SVR can outperform purely linear baselines such as ordinary least squares or traditional statistical time-series models. This observation is consistent with results reported by Zhao et al., who compare statistical forecasting techniques, linear regression, and kernel-based models and find that SVR provides improved predictive accuracy relative to linear or purely autoregressive approaches Zhao et al. (2020).

Collectively, these studies indicate that classical ML approaches—particularly tree-based ensembles and support vector regression—represent strong baselines for parking availability forecasting. Their advantages include relatively low computational requirements, robustness to moderate dataset sizes, and a degree of interpretability that facilitates model analysis. Importantly, several comparative evaluations report that these models achieve predictive performance comparable to or exceeding that of more complex deep learning architectures, especially in short-horizon forecasting scenarios and facility-level datasets. These findings challenge the assumption that increasing model complexity necessarily leads to improved predictive accuracy.

2.3.2 Deep Learning and Hybrid Neural Approaches

Deep learning (DL) models have increasingly been applied to parking availability forecasting due to their capacity to model complex temporal dependencies directly from sequential data. Recurrent neural networks (RNN), gated recurrent units (GRU), and long short-term memory (LSTM) architectures are particularly common, as they are specifically designed to capture long-range temporal relationships in time-series data while mitigating vanishing gradient problems.

Domenech et al. investigate several recurrent neural architectures—including RNN, GRU, and LSTM—for urban parking prediction and demonstrate that sequence-based models can outperform traditional machine learning baselines when sufficient historical context is available Domenech et al. (2025). Their work highlights the importance of modeling temporal regularities and nonlinear dynamics in urban parking demand. Similarly, Bilotta et al. focus on multi-step forecasting scenarios by predicting parking availability over a 24-hour horizon with fine temporal resolution Bilotta et al. (2023). Using recurrent neural architectures enriched with traffic and weather features, their study shows that deep learning models can provide improved accuracy for longer forecasting horizons where simpler autoregressive models tend to degrade.

Beyond purely temporal models, several studies have introduced spatiotemporal deep learning architectures that incorporate spatial dependencies between

parking facilities. Yang et al. propose a graph convolutional neural network combined with LSTM layers (GCNN-LSTM), enabling the model to capture both spatial correlations and temporal dynamics in parking availability Yang et al. (2019). Their results demonstrate that incorporating spatial relationships between parking locations can significantly improve predictive accuracy in dense urban environments where parking demand exhibits spatial spillover effects.

Hybrid neural architectures represent another emerging direction in the literature. Elomiya et al. introduce a hybrid forecasting framework combining adaptive neuro-fuzzy inference systems (ANFIS) with recurrent neural networks Elomiya et al. (2024). By integrating fuzzy logic with neural sequence modeling, the authors aim to enhance prediction robustness under noisy sensor conditions and improve resilience to uncertainty in the input data. Such hybrid approaches reflect broader efforts to address the volatility and non-stationarity often observed in parking availability time series.

Despite the increasing adoption of deep learning methods, comparative studies consistently indicate that these models do not universally outperform classical machine learning approaches. Several empirical evaluations report that while deep architectures can excel in long-horizon or spatiotemporal forecasting tasks, ensemble-based machine learning models often remain competitive or superior in short-term forecasting settings with structured feature inputs Awan et al. (2020); Channamallu et al. (2023c); Koumetio Tekouabou et al. (2022). In addition, Vlahogianni et al. emphasize that practical deployment considerations—such as computational efficiency, real-time responsiveness, and system integration—must be considered alongside predictive accuracy when selecting forecasting models for operational parking systems Vlahogianni et al. (2016).

Overall, the literature demonstrates that both classical machine learning and deep learning models contribute valuable modeling capabilities for parking availability forecasting. However, empirical evidence consistently suggests that the relative performance of these approaches depends strongly on dataset characteristics, feature engineering strategies, and prediction horizon. Consequently, no single modeling paradigm consistently dominates across all scenarios, highlighting the importance of systematic comparative evaluation and horizon-aware analysis when designing practical parking forecasting systems Awan et al. (2020); Vlahogianni et al. (2016); Channamallu et al. (2023c); Koumetio Tekouabou et al. (2022); Bilotta et al. (2023).

2.4 Research Gaps and Opportunities

The review of existing literature demonstrates significant progress in parking availability forecasting, particularly with respect to data-driven modeling approaches and the increasing availability of sensor-based datasets. However, several important limitations remain that motivate further investigation. These gaps relate primarily to forecasting horizon analysis, model selection strategies, operational deployment, and interpretability of predictive models.

First, while many studies evaluate multiple forecasting algorithms, relatively few analyze how predictive performance varies systematically across prediction horizons. Most comparative evaluations report aggregate performance metrics across all forecast steps, implicitly assuming that a single model can provide optimal predictions for all lead times Awan et al. (2020); Channamallu et al.

(2023c). However, studies that examine multi-step forecasting more closely indicate that model accuracy typically deteriorates as the prediction horizon increases, and that different model architectures may perform better at different lead times Vlahogianni et al. (2016); Bilotta et al. (2023). Despite these observations, the literature rarely investigates horizon-dependent performance explicitly or incorporates this variability into model design. This limitation suggests a need for forecasting frameworks that explicitly account for horizon-specific model strengths.

Second, the majority of parking forecasting studies rely on single-model approaches, even when multiple candidate models are evaluated during experimentation. Although ensemble methods such as random forests or boosting models demonstrate strong performance in structured parking datasets Awan et al. (2020); Koumetio Tekouabou et al. (2022), and deep learning models have shown advantages in sequential and multi-step prediction tasks Domenech et al. (2025); Bilotta et al. (2023), relatively little work explores systematic hybridization strategies that combine complementary models. In particular, few studies propose mechanisms for dynamically selecting or blending models depending on forecast horizon or data characteristics. This indicates an opportunity to investigate hybrid forecasting frameworks that leverage the strengths of different model families within a unified prediction system.

Third, much of the existing literature evaluates predictive models in offline experimental environments without addressing their integration into operational systems. Many studies focus primarily on algorithmic accuracy using historical datasets, while practical considerations such as automated data ingestion, real-time model inference, and system scalability receive comparatively little attention Vlahogianni et al. (2016). As a result, the transition from experimental forecasting models to deployable decision-support systems remains insufficiently explored. Developing architectures that integrate predictive models into real-time data pipelines therefore represents an important research direction for smart city applications.

A further limitation concerns the translation of predictive outputs into actionable decision-support tools. Although parking forecasting research often emphasizes statistical accuracy metrics such as RMSE or MAE, fewer studies investigate how predictions can be communicated to end users in a way that supports practical decision making. For drivers, city planners, and parking operators, the usefulness of forecasting systems depends not only on prediction accuracy but also on how clearly future availability information can be visualized and interpreted within operational contexts. Consequently, integrating forecasting models with user-oriented visualization tools and interactive dashboards remains an underdeveloped area in the literature.

Finally, interpretability and feature-level insights remain relatively underexplored in parking availability forecasting research. Many recent studies prioritize predictive performance, particularly when applying deep learning architectures, but provide limited analysis of which input variables contribute most strongly to model predictions. Understanding the relative importance of temporal, contextual, and environmental variables is valuable both for improving model design and for supporting transparency in data-driven decision systems. Feature importance analysis can therefore play an important role in bridging predictive modeling with practical insights about parking demand dynamics.

Addressing these gaps requires a more integrated perspective that combines methodological rigor with practical system design. In response, this thesis adopts a horizon-aware forecasting framework that systematically evaluates model performance across prediction horizons and introduces hybrid model blending strategies to leverage complementary strengths of different algorithms. Furthermore, the proposed approach integrates predictive models within a real-time data engineering pipeline and presents the resulting forecasts through an interactive dashboard designed to support decision making in urban parking contexts. Through this combination of methodological analysis and operational implementation, the thesis contributes to both the academic literature on parking availability forecasting and the practical development of data-driven parking management systems.

Chapter 3

Method

This chapter presents the methodological framework underlying the development, evaluation, and deployment of the proposed parking availability forecasting system. In contrast to purely theoretical or simulation-based studies, this work adopts an end-to-end implementation perspective that integrates data engineering, predictive modeling, and real-time system deployment into a unified pipeline.

As illustrated in Fig. 3.1, the proposed methodology is structured into three main layers: the data layer, the modeling layer, and the system and application layer. The process begins with the collection of parking availability data and external weather information, which are transformed through aggregation and feature engineering steps to construct a structured dataset. In the modeling layer, multiple baseline machine learning and deep learning models are trained and evaluated, followed by the development of a horizon-aware hybrid model that combines their strengths. Additionally, feature importance analysis is performed to identify the most influential variables. Finally, the system and application layer operationalizes the selected model within a deployment pipeline, enabling continuous prediction and integration into a decision support dashboard that provides short-term availability forecasts.

The chapter is structured in three main parts. First, the data sources and preprocessing procedures are described in detail, including data structure, feature engineering, and temporal aggregation. Second, the implementation of predictive models is presented, covering baseline approaches and the proposed horizon-aware hybrid model, along with the evaluation strategy across forecasting horizons. Finally, the design of the real-time deployment pipeline is explained, including data ingestion, model inference, storage of predictions, and dashboard integration.

Together, these components establish a coherent and deployment-ready methodology that connects data processing, horizon-aware forecasting, and real-time visualization into an integrated system.

3.1 Dataset and Feature Engineering

This section describes the data sources and feature engineering steps used to prepare the input for the forecasting models. It first introduces the parking avail-

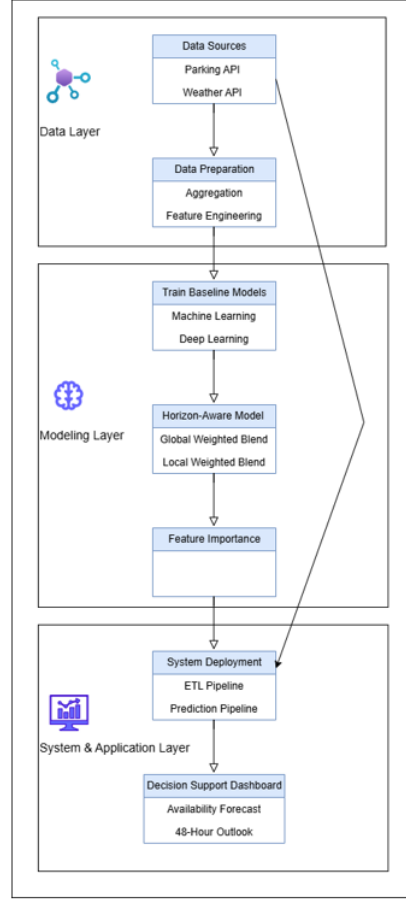


Figure 3.1: Overview of the proposed real-time deployment architecture.

ability data and its structure, followed by the integration of external weather information. Next, an exploratory data analysis is conducted to identify temporal patterns and underlying trends in the data. Finally, the preprocessing and feature construction steps are presented, including data aggregation, transformation, and the creation of time-based and statistical features required for model training.

3.1.1 Parking Availability Data

The parking availability data used in this study is provided by the project partner *Smart City System Parking Solutions GmbH* and accessed via the Parking-Pilot API, which offers historical parking records in CSV format. The analysis focuses on two off-street park-and-ride (P+R) facilities located in the city of Bamberg, Germany. These facilities were selected due to reliable sensor coverage, continuous data availability, and their operational relevance for commuter-oriented parking demand.

The first facility, **P+R Brennerstraße (ID = 38)**, is located at Brennerstraße 9a, 96052 Bamberg, adjacent to the central train station. Owing to

its proximity to public transportation infrastructure, this facility exhibits pronounced commuter-driven usage patterns. The total capacity comprises 113 parking spaces

The second facility, **P+R Heinrichsdamm (ID = 634)**, is located at Rhein-Main-Donau-Damm, 96047 Bamberg. The total capacity of this facility is 800 parking spaces. Of these, 64 outdoor spaces are equipped with individual parking sensors and are therefore included in the analysis.

For both facilities, the ParkingPilot API provides space-level parking events describing vehicle arrivals and departures. Each record includes the parking lot identifier, parking space identifier, arrival timestamp, departure timestamp, and corresponding human-readable time representations. Although the raw data consists of event-level records, these events enable reconstruction of *Parking Availability*, defined in this study as the number of free parking spaces at a given point in time.

The dataset covers the period from 30 November 2024 to 28 November 2025, providing nearly one year of continuous observations. This time span captures daily, weekly, and seasonal patterns in parking availability, including commuter cycles and holiday-related variations. All timestamps are provided in a consistent time reference to ensure temporal comparability across facilities and compatibility with external data sources.

Data retrieval is fully automated through programmatic API access. The raw event-level data is persisted in a PostgreSQL relational database within the Parking database instance. The records are stored in the public schema under the table `ali_parking_operations_2025.11.29`.

The table contains the following columns: `parking_space_id`, `arrival_unix_seconds`, `departure_unix_seconds`, `arrival_unix_seconds_humanreadable`, and `departure_unix_seconds_humanreadable`.

An excerpt of the stored raw data is illustrated in Figure 3.2. Persisting the original event-level records without modification ensures full traceability, reproducibility, and data integrity, forming the foundational data layer for all subsequent preprocessing and modeling steps.

<code>parking_lot_id</code> [PK] integer	<code>parking_space_id</code> [PK] integer	<code>arrival_unix_seconds</code> [PK] bigint	<code>departure_unix_seconds</code> bigint	<code>arrival_unix_seconds_humanreadable</code> timestamp with time zone	<code>departure_unix_seconds_humanreadable</code> timestamp with time zone
38	643	1733087724	1733089834	2024-12-01 21:15:24+00	2024-12-01 21:50:34+00
38	603	1733079618	1733104430	2024-12-01 19:00:18+00	2024-12-02 01:53:50+00
38	644	1733079576	1733085056	2024-12-01 18:59:36+00	2024-12-01 20:30:56+00
38	608	1733072858	1733129494	2024-12-01 17:07:38+00	2024-12-02 08:51:34+00

Figure 3.2: Raw parking event data (PostgreSQL excerpt).

3.1.2 Weather Data

To account for external factors that may influence parking availability, historical weather data is collected for the same geographic area and observation period. Meteorological observations are retrieved from the Open-Meteo historical weather service, which provides publicly accessible hourly weather data.

The weather dataset includes hourly measurements of air temperature (°C), relative humidity (%), and precipitation (mm) for the city of Bamberg. Weather data is collected for the identical time span as the parking data, from 30 Novem-

ber 2024 to 28 November 2025, ensuring full temporal alignment between both data sources.

All weather records are stored with standardized timestamps (timestamp with time zone) to ensure precise temporal alignment with the parking availability data. The weather observations are persisted in the PostgreSQL Parking database within the public schema under the table `ali_weather_bamberg_hourly`.

The table contains the columns `timestamp`, `temperature_2m`, `relative_humidity_2m`, and `precipitation`. Similar to the parking event data, weather observations are ingested automatically and stored in their original, unmodified form. This guarantees consistency, traceability, and reproducibility of the preprocessing and modeling procedures described in the subsequent sections. An excerpt of the stored weather data is illustrated in Figure 3.3.

<code>timestamp</code> [PK] timestamp with time zone	<code>temperature_2m</code> double precision	<code>relative_humidity_2m</code> double precision	<code>precipitation</code> double precision
2024-01-01 00:00:00+00	4	87	0
2024-01-01 01:00:00+00	4.5	89	0
2024-01-01 02:00:00+00	4.7	87	0
2024-01-01 03:00:00+00	4.5	87	0.1

Figure 3.3: Hourly weather data (PostgreSQL excerpt).

3.1.3 Exploratory Data Analysis

Hourly Parking Availability Patterns

To better understand the temporal structure of parking availability, the average number of free spaces was analyzed as a function of the hour of day for both parking facilities. Figure 3.4 presents the mean hourly availability together with the interquartile range (IQR, 25th–75th percentile), illustrating the variability of availability across days.

A clear diurnal pattern is visible in both parking facilities. During the night and early morning hours (approximately 00:00–05:00), the number of free spaces remains relatively high and stable, reflecting low overnight demand. Beginning around 05:00, availability gradually decreases as commuting activity increases. The lowest availability occurs between approximately 09:00 and 13:00, indicating the peak parking demand period for both facilities.

After early afternoon, the trend reverses and availability gradually increases as parking demand declines. By the evening hours, the number of free spaces approaches the nighttime baseline level.

Although both facilities exhibit a similar temporal structure, differences in magnitude and variability can be observed. Lot 634 shows a deeper midday reduction and wider interquartile range during peak hours, indicating stronger demand fluctuations. In contrast, Lot 38 demonstrates a smoother decline and recovery pattern.

These consistent diurnal patterns confirm that parking availability is strongly time-dependent and justify the inclusion of hour-of-day features in the forecasting models. In addition, the increased variability during peak periods suggests

3.1. DATASET AND FEATURE ENGINEERING

higher uncertainty during midday demand, which is relevant when evaluating forecasting performance across prediction horizons.

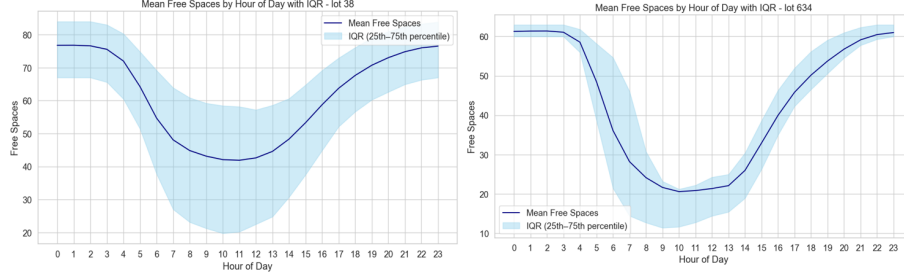


Figure 3.4: Mean hourly parking availability for Lot 38 and Lot 634 with interquartile range.

Weekly Parking Availability Patterns

To analyze weekly demand variations, the average number of free parking spaces was examined for each day of the week for both parking facilities. Figure 3.5 illustrates the mean availability together with variability across days.

The results indicate noticeable differences between weekdays and weekends. For both parking facilities, Sunday shows the highest average number of free spaces, indicating comparatively low parking demand. This pattern likely reflects reduced commuter activity and lower overall mobility compared to weekdays. In contrast, weekdays generally exhibit lower availability levels, suggesting higher parking demand associated with work-related trips and daily urban activities.

Across the weekday period, availability remains relatively stable, although minor fluctuations can be observed. For Lot 38, the number of free spaces is slightly lower between Monday and Friday compared to the weekend, indicating consistent utilization throughout the working week. A similar trend is visible for Lot 634, although this facility exhibits larger variability across days, as reflected in the wider error bars.

The increased variability during certain weekdays suggests that parking demand may be influenced by factors such as work schedules, local events, or varying traffic conditions. Nevertheless, the overall weekly pattern remains relatively consistent, with reduced availability during the working week and increased availability during the weekend.

These findings suggest that weekly temporal patterns are present in the dataset and support the inclusion of day-of-week features in the forecasting models. Capturing such weekly cycles can improve predictive performance by enabling models to account for recurring demand patterns associated with commuting and weekend activity.

Monthly Parking Availability Patterns

To examine potential seasonal effects in parking demand, the average number of free parking spaces was analyzed for each month of the year. Figure 3.6 presents the mean monthly availability for both parking facilities.

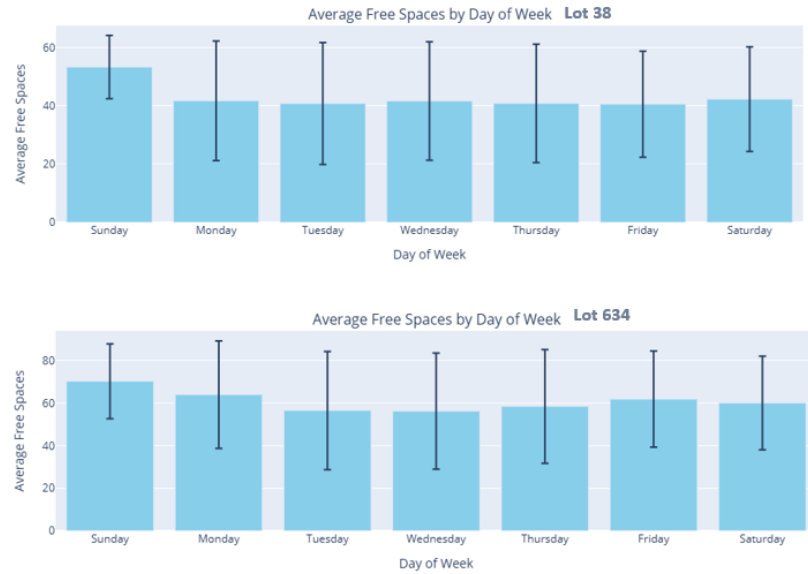


Figure 3.5: Average parking availability by weekday for Lot 38 and Lot 634 (bars: mean, error bars: variability).

The results indicate moderate seasonal variation in parking availability throughout the year. For Lot 38, availability is relatively higher during the winter months, particularly in January, and again increases noticeably during late summer and early autumn. The highest average availability occurs around September, suggesting reduced parking demand during this period. In contrast, the lowest availability levels are observed during the spring months, especially between March and May, indicating relatively higher parking utilization during this time.

For Lot 634, the seasonal pattern appears less pronounced, with availability levels remaining relatively stable across most months. Minor variations can still be observed, with slightly higher availability during the early summer months and somewhat lower values during the spring period. Compared to Lot 38, the magnitude of monthly fluctuations is smaller, suggesting that demand at this facility is more stable throughout the year.

Overall, the monthly patterns indicate the presence of mild seasonal effects in parking availability. Such variations may be influenced by factors including tourism activity, seasonal mobility patterns, or local events. While these monthly differences are less pronounced than the daily or hourly patterns observed earlier, they still provide useful contextual information about long-term demand trends.

These observations support the inclusion of month-related temporal features in the forecasting models, allowing the models to capture seasonal demand variations that may influence parking availability over longer time horizons.

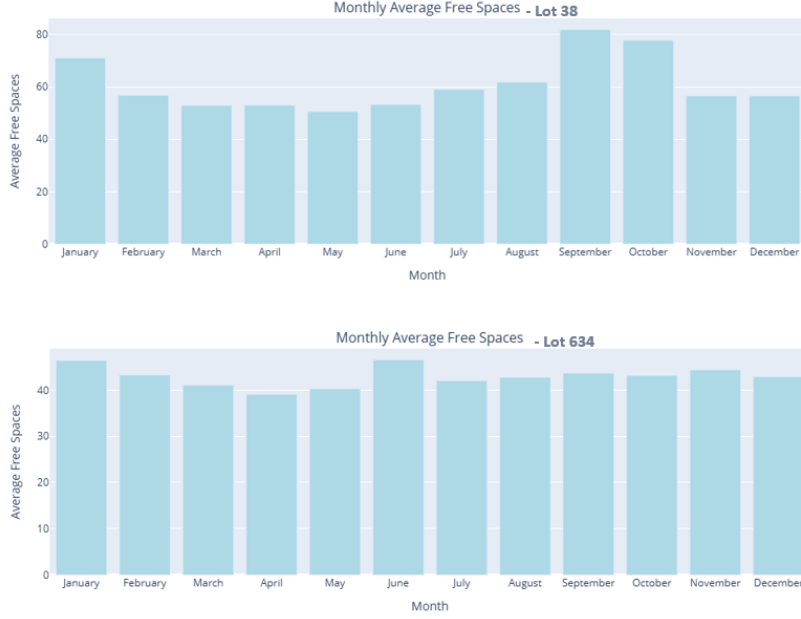


Figure 3.6: Average parking availability by month for Lot 38 and Lot 634.

3.1.4 Data Preprocessing and Feature Construction

Following data ingestion, the raw parking event data is transformed into a structured time series representation suitable for predictive modeling. Preprocessing is applied consistently to both parking facilities (Lot 38 and Lot 634) and comprises the construction of a continuous time-indexed signal from event-based records. In a first step, space-level arrival and departure events are converted into minute-level counts of occupied and free parking spaces. Subsequent steps aggregate the minute-level signal to the modeling resolution and integrate exogenous weather observations; these steps are described in the following subsections.

Minute-level construction of parking availability

The ParkingPilot API provides space-level parking events that describe vehicle arrivals and departures for each sensor-instrumented parking space. Each event i is represented by an interval $[a_i, d_i)$, where a_i denotes the arrival timestamp and d_i the departure timestamp. To obtain a continuous time series, a minute-level time axis t is generated for the full observation window for each parking lot l .

At each minute t , the number of occupied spaces is computed by counting how many parking-event intervals cover t :

$$occupied_spaces_l(t) = \sum_{i=1}^{N_l} \mathbb{I}(a_i \leq t < d_i), \quad (3.1)$$

where N_l is the number of recorded parking events for lot l and $\mathbb{I}(\cdot)$ denotes the indicator function. Parking availability is defined in this study as the number of free parking spaces. Let C_l denote the sensor-covered capacity of lot l (i.e., the number of monitored spaces). The minute-level parking availability is then derived as:

$$free_spaces_l(t) = C_l - occupied_spaces_l(t). \quad (3.2)$$

This transformation converts the event-based records into a uniform minute-resolution parking availability time series for each facility. The resulting dataset contains the fields `minute_timestamp` (timestamp with time zone), `parking_lot_id`, `free_spaces`, and `occupied_spaces`.

The minute-level availability data is persisted in PostgreSQL in separate tables for each parking facility, namely `ali_parking_availability_minute_lot38` and `ali_parking_availability_minute_lot634`. Storing each lot independently ensures clear separation of facility-level signals and supports modular preprocessing and modeling.

Hourly Aggregation

To align the target representation with the short-term forecasting objective and to reduce high-frequency minute-level fluctuations, the minute-resolution parking availability series is aggregated to an hourly resolution. Each `minute_timestamp` is mapped to a corresponding `hour_timestamp` by truncating to the beginning of the respective hour.

For each parking facility, hourly availability statistics are computed based on the underlying minute-level `free_spaces` signal. Specifically, the following summary measures are derived for every hour and parking lot:

- `avg_free_spaces`: average number of free spaces within the hour,
- `min_free_spaces`: minimum number of free spaces observed within the hour,
- `max_free_spaces`: maximum number of free spaces observed within the hour,
- `free_spaces_range`: difference between maximum and minimum availability within the hour.

The average value (`avg_free_spaces`) serves as the primary forecasting target, while the additional measures describe intra-hour variability and short-term volatility.

In addition, calendar-based temporal descriptors are extracted from the `hour_timestamp` to capture recurring temporal patterns. These include `hour_of_day`, `day_of_week`, `weekday_name`, `day_of_month`, `calendar_week`, `month_number`, and `month_name`.

The aggregated hourly parking availability data is stored in PostgreSQL in separate tables for each facility, namely `ali_parking_availability_hourly_lot38` and `ali_parking_availability_hourly_lot634`. An excerpt of the resulting hourly dataset is shown in Figure 3.7.

3.1. DATASET AND FEATURE ENGINEERING

	hour_timestamp timestamp with time zone	parking_lot_id integer	avg_free_spaces numeric	min_free_spaces integer	max_free_spaces integer	free_spaces_range numeric	hour_of_day double precision	day_of_week double precision	weekday_name text	day_of_month double precision	calendar_week double precision	month_number double precision	month_name text
1	2024-10-12 00:00:00+00	634	64.00	64	64	0.00	0	6	Saturday	12	41	10	October
2	2024-10-12 01:00:00+00	634	64.00	64	64	0.00	1	6	Saturday	12	41	10	October
3	2024-10-12 02:00:00+00	634	64.00	64	64	0.00	2	6	Saturday	12	41	10	October
4	2024-10-12 03:00:00+00	634	64.00	64	64	0.00	3	6	Saturday	12	41	10	October
5	2024-10-12 04:00:00+00	634	64.00	64	64	0.00	4	6	Saturday	12	41	10	October

Figure 3.7: Hourly parking availability (Lot 634) stored in PostgreSQL.

Integration with Weather Observations

The hourly parking availability dataset is subsequently integrated with hourly weather observations for the city of Bamberg. The integration is performed via an inner join on the corresponding hour_timestamp, ensuring precise temporal alignment between parking availability and meteorological conditions.

The incorporated weather variables include temperature_2m, relative_humidity_2m, and precipitation. These exogenous features provide additional contextual information that may influence short-term parking availability patterns.

In addition to the numerical weather variables, categorical context features are derived to capture coarse behavioral regimes that support stable pattern learning. These include day_type (weekday versus weekend), time_period (e.g., rush hours versus nighttime), temperature_category, and precipitation_category. The categorical abstractions are designed to complement the continuous weather measurements by providing regime-level information that may enhance model robustness under varying environmental conditions.

The resulting unified hourly dataset therefore combines parking availability statistics, calendar descriptors, and weather-related features into a single modeling table. For reproducibility and modular system design, the integrated training datasets are stored separately for each facility in PostgreSQL under the tables ali_training_data_hourly_availability_hourly_lot38 and

ali_training_data_hourly_availability_hourly_lot634.

An excerpt of the integrated hourly training dataset is shown in Figure 3.8.

hour_timestamp timestamp with time zone	parking_lot_id integer	avg_free_spaces numeric	min_free_spaces integer	max_free_spaces integer	free_spaces_range numeric	hour_of_day double precision	day_of_week double precision	weekday_name text	day_of_month double precision	calendar_week double precision	month_number double precision	month_name text	temperature_2m double precision	relative_humidity_2m double precision	precipitation double precision	day_type text	time_period text	temperature_category text	precipitation_category text
2024-10-12 00:00:00+00	634	64.00	64	64	0.00	0	6	Saturday	12	41	10	October	5.0	90	0	Weekend	Night	Cool	NoRain
2024-10-12 01:00:00+00	634	64.00	64	64	0.00	1	6	Saturday	12	41	10	October	4.5	90	0	Weekend	Night	Cool	NoRain
2024-10-12 02:00:00+00	634	64.00	64	64	0.00	2	6	Saturday	12	41	10	October	4.2	90	0	Weekend	Night	Cool	NoRain
2024-10-12 03:00:00+00	634	64.00	64	64	0.00	3	6	Saturday	12	41	10	October	4.0	90	0	Weekend	Night	Cool	NoRain
2024-10-12 04:00:00+00	634	64.00	64	64	0.00	4	6	Saturday	12	41	10	October	3.8	90	0	Weekend	Night	Cool	NoRain

Figure 3.8: Integrated training dataset (hourly).

Feature Engineering and Model Input Construction

After constructing the unified hourly dataset (parking availability, weather observations, and calendar descriptors), additional feature engineering is performed in Python prior to model training. This step transforms the integrated data into the final model-ready input table and introduces short-term dynamics and local trend representations that are not directly available from the raw hourly aggregates.

To capture short-term temporal structure and recent availability behavior, rolling window statistics are computed over a 24-hour horizon. Specifically, a 24-hour rolling mean (avg_free_spaces_rolling_24h) and rolling standard deviation (avg_free_spaces_std_24h) are derived from the avg_free_spaces signal. These features summarize local central tendency and variability, enabling the models to detect regime stability and short-term fluctuations.

In addition, first-order differences are calculated to represent hour-to-hour changes. The feature `avg_free_spaces_diff_1h` captures the immediate change in parking availability between consecutive hours, while `temperature_diff_1h` reflects short-term meteorological variation. These differenced features help the models learn directional movement rather than relying solely on absolute levels.

Categorical context variables (`day_type`, `time_period`, `temperature_category`, `precipitation_category`) are transformed into ordinal numeric representations to ensure compatibility with classical machine learning algorithms while preserving regime-level structure.

The resulting final hourly training dataset contains the following model input variables:

- `avg_free_spaces`,
- `min_free_spaces`,
- `max_free_spaces`,
- `free_spaces_range`,
- `temperature_2m`,
- `relative_humidity_2m`,
- `precipitation`,
- `hour_of_day`,
- `day_of_week`,
- `day_of_month`,
- `calendar_week`,
- `month`,
- `day_type`,
- `time_period`,
- `temperature_category`,
- `precipitation_category`,
- `avg_free_spaces_rolling_24h`,
- `avg_free_spaces_std_24h`,
- `avg_free_spaces_diff_1h`,
- `temperature_diff_1h`.

The primary forecasting target is `avg_free_spaces`. For one-step-ahead and multi-step forecasting, the most recent observed availability level is also retained among the explanatory variables to provide autoregressive information to the models.

After completing feature engineering, the hourly dataset is transformed into supervised learning samples and split chronologically for reproducible model evaluation. To prevent temporal leakage and ensure realistic forecasting conditions, the data is divided strictly in time order. Let T denote the total number of hourly observations after cleaning and feature construction. The dataset is partitioned into 70% training, 15% validation, and 15% testing data by defining split indices at $0.70T$ and $0.85T$. The earliest segment is used for model training, the subsequent segment for validation and hyperparameter tuning, and the most recent segment for final out-of-sample testing.

To support the decision-support objective of forecasting the next 24 hours of parking availability, the time series is converted into a supervised learning problem using a sliding-window formulation. Let $X_t \in \mathbb{R}^p$ denote the vector of engineered input features at hour t , where $p = 20$, and let y_t denote the target variable corresponding to avg_free_spaces at hour t .

For each training example starting at index k , the model input consists of the previous 48 hours of multivariate observations:

$$X^{(k)} = [X_{k-48}, X_{k-47}, \dots, X_{k-1}] \in \mathbb{R}^{48 \times p},$$

while the prediction target corresponds to the next 24-hour availability trajectory:

$$y^{(k)} = [y_k, y_{k+1}, \dots, y_{k+23}] \in \mathbb{R}^{24}.$$

This sliding-window formulation produces supervised datasets for training, validation, and testing with the following tensor shapes (number of samples, input sequence length, number of features):

- x_{train} : (11695, 48, 20), y_{train} : (11695, 24)
- x_{val} : (2468, 48, 20), y_{val} : (2468, 24)
- x_{test} : (2445, 48, 20), y_{test} : (2445, 24)

For classical machine-learning models that require two-dimensional input representations, each 48×20 input window is flattened into a single feature vector per sample, while the 24-dimensional output remains in multi-step form via multi-output regression. Deep learning models, in contrast, operate directly on the three-dimensional input tensors.

Finally, all numerical input features are normalized using min-max scaling computed exclusively on the training set. The same scaling parameters are subsequently applied to the validation and test sets to maintain comparability and avoid information leakage.

3.2 Model Development and Evaluation

To identify a robust forecasting approach for next-day parking availability, a comprehensive set of predictive models was implemented and systematically evaluated. The experimental design includes both classical machine learning regressors and deep learning sequence architectures. This dual perspective enables a structured comparison between interpretable, computationally efficient baseline models and more expressive nonlinear architectures capable of capturing complex temporal dependencies and multi-horizon dynamics.

Table 3.1: Predictive models and key hyperparameters.

Model	Key hyperparameters
Linear / Ridge Regression	Multi-output regression; Ridge: $\alpha = 1.0$.
Decision Tree	Max depth = 20; min samples split = 10; min leaf = 5.
Random Forest	100 trees; max depth = 20; min samples split = 10; min leaf = 5.
Gradient Boosting	50 estimators; depth = 5; learning rate = 0.1.
XGBoost	100 estimators; depth = 5; learning rate = 0.1.
SVR (RBF)	$C = 1.0$; $\epsilon = 0.1$; max 5,000 samples.
RNN-based models	LSTM / GRU / BiLSTM: two layers (150, 75 units).
Encoder-Decoder	LSTM encoder-decoder; 24-step output horizon.
CNN-RNN hybrids	1D CNN (64 filters, $k = 3$) + RNN layers (150, 75).

All models were trained on the supervised learning dataset $(\mathbf{X}_{train}, \mathbf{y}_{train})$ derived from the preprocessing and feature engineering pipeline described in the previous section. Model development, validation, and testing were conducted under a consistent chronological evaluation protocol to ensure comparability and to prevent temporal leakage. The same modeling framework and training procedure were applied independently to both study sites (Lot 38 and Lot 634), enabling a controlled cross-facility comparison.

Hyperparameter selection followed a structured tuning strategy combining insights from prior literature with validation-based empirical optimization. The finalized hyperparameter configurations for each model are summarized in Table 3.1.

3.2.1 Classical Machine Learning Models

Classical machine learning models were implemented as computationally efficient and interpretable baselines for multi-horizon parking availability forecasting. These models operate on structured tabular feature representations and provide strong performance under limited computational cost, making them suitable for comparison against more complex deep learning architectures.

To cover complementary modeling assumptions, a diverse set of linear, kernel-based, and tree-based regressors was evaluated. Linear Regression was included as a transparent baseline model assuming an approximately linear relationship between explanatory variables and parking availability. Ridge Regression extends this formulation by introducing ℓ_2 regularization to reduce sensitivity to multicollinearity and improve generalization stability Hoerl and Kennard (1970).

To capture nonlinear relationships and feature interactions, tree-based methods were incorporated. A Decision Tree Regressor partitions the feature space recursively into locally homogeneous regions Loh (2011). Random Forest extends this concept through bootstrap aggregation of decorrelated trees, reducing variance and improving robustness Breiman (2001). Gradient Boosting meth-

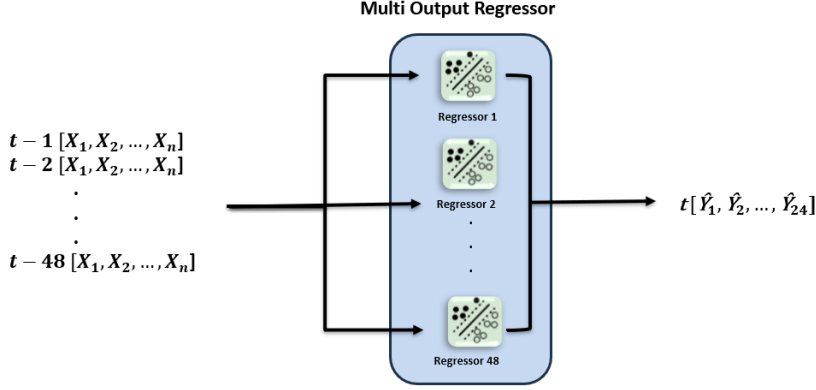


Figure 3.9: Multi-output forecasting: 48 past time steps mapped to 24 future predictions.

ods construct an additive model by sequentially fitting weak learners to residual errors Friedman, Jerome H. (2001). XGBoost was evaluated as an optimized gradient-boosted tree implementation incorporating regularization, shrinkage, and computational efficiency enhancements Chen and Guestrin (2016).

As a complementary nonlinear approach, Support Vector Regression (SVR) with a radial basis function kernel was also evaluated. SVR formulates regression as a margin-based optimization problem in a high-dimensional feature space, providing strong generalization performance under nonlinear settings Smola and Schölkopf (2004). Due to its computational complexity, SVR was trained on a capped subset of 5,000 samples, while evaluation was performed consistently on the full validation and test sets.

Multi-output Forecasting Formulation

The forecasting task requires predicting parking availability for the next 24 hours simultaneously. Let $X \in \mathbb{R}^{N \times d}$ denote the flattened input matrix constructed from the previous 48 hours of observations, and let $Y \in \mathbb{R}^{N \times 24}$ denote the corresponding multi-step target matrix, where each row represents the 24-hour future availability trajectory.

As illustrated in Fig. 3.9, the multi-output forecasting setup maps a sequence of past observations (48 time steps with d features each) to a vector of 24 future predictions. In this formulation, each output dimension corresponds to a specific forecast horizon (e.g., $t + 1$ to $t + 24$).

Models with native multi-output capability were trained directly on (X, Y) . For inherently single-output estimators, a multi-output wrapper strategy was employed, where one independent regressor is trained for each forecast horizon. This results in a set of parallel models, each specialized for a specific lead time, as depicted in the figure.

At inference time, the 24 horizon-specific predictions are concatenated to form the complete next-day availability forecast, ensuring alignment with the decision-support objective.

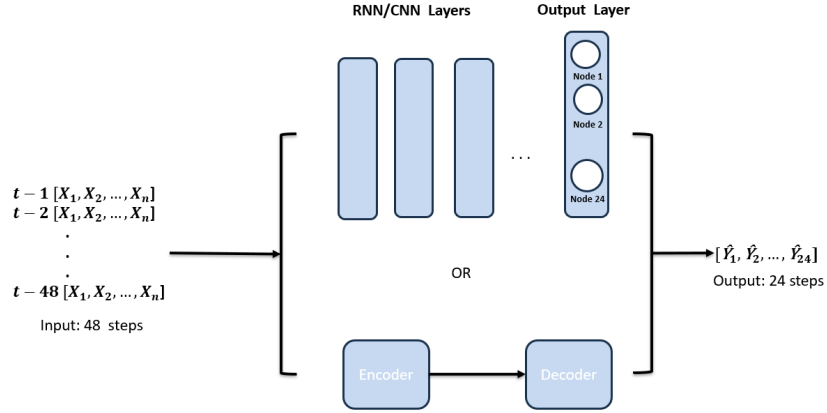


Figure 3.10: Deep learning forecasting architectures: sequence models (RNN/CNN) and encoder–decoder for direct multi-step prediction.

3.2.2 Deep Learning Models

To complement the classical machine learning baselines, several deep learning architectures were implemented to explicitly model temporal dependencies in multivariate time series. In contrast to classical regressors that operate on flattened feature vectors, deep models preserve the sequential structure of the input data. Each training sample is represented as a tensor $X \in \mathbb{R}^{48 \times F}$, corresponding to the previous 48 hours of engineered features (with $F = 20$), and the model outputs a 24-dimensional vector $Y \in \mathbb{R}^{24}$ representing the predicted parking availability for the next 24 hours.

As illustrated in Fig. 3.10, two main deep learning paradigms were considered. In the first approach, sequence models such as recurrent neural networks (RNNs) and convolutional neural networks (CNNs) process the input sequence directly through stacked temporal layers. The learned representation is then passed to a fully connected output layer with 24 neurons, where each neuron corresponds to a specific forecast horizon. This setup enables the model to capture temporal patterns while producing all future predictions simultaneously.

In the second approach, an encoder–decoder architecture is employed. The encoder processes the input sequence and compresses the temporal information into a latent representation, which is then passed to the decoder. The decoder generates the full sequence of future predictions, effectively learning the mapping from past observations to future trajectories. This structure is particularly suitable for sequence-to-sequence forecasting tasks with longer horizons.

Importantly, both architectures follow a direct multi-step, non-recursive forecasting strategy. Instead of predicting one step ahead and feeding predictions back into the model, all 24 future values are predicted simultaneously in a single forward pass. This avoids error accumulation across time steps and ensures stable performance across longer forecast horizons.

Recurrent Architectures

Recurrent neural networks (RNNs) were selected as the primary deep learning family due to their ability to maintain an internal state and model temporal dependencies across time steps. Standard RNNs suffer from vanishing gradient issues in longer sequences; therefore, gated recurrent architectures were employed.

Long Short-Term Memory (LSTM) networks introduce input, forget, and output gates that regulate information flow and enable learning of longer-range temporal dependencies Hochreiter and Schmidhuber (1997). Gated Recurrent Units (GRU) provide a simplified gating mechanism with fewer parameters while retaining strong modeling capacity Cho et al. (2014). Both architectures were evaluated using stacked recurrent layers followed by fully connected output layers producing the 24-hour forecast.

In addition, bidirectional LSTM models were implemented. Although forecasting is inherently forward-looking, bidirectional processing can improve representation learning within the historical input window by allowing the model to exploit full context when encoding the past sequence Schuster and Paliwal (1997). The resulting encoded representation is then mapped to the multi-step output layer.

Encoder–Decoder (Sequence-to-Sequence) Architecture

To explicitly model the mapping from historical context to future trajectories, an encoder–decoder architecture was implemented. In this formulation, an LSTM encoder compresses the 48-hour input sequence into a latent representation vector. A decoder network then generates the 24-hour forecast horizon from this representation. This sequence-to-sequence design is particularly suitable for multi-step forecasting, as it learns a direct transformation from past observations to a structured future sequence Sutskever et al. (2014).

Hybrid Convolutional–Recurrent Models

Beyond purely recurrent models, hybrid architectures combining convolutional and recurrent components were evaluated. In these models, one-dimensional convolutional layers are first applied along the temporal axis to extract short-range temporal patterns and local feature interactions. The resulting feature maps are then passed to recurrent layers (LSTM, GRU, or Bidirectional LSTM) that model longer-range dependencies.

This architecture leverages the ability of convolutional layers to capture localized temporal motifs while retaining the sequence modeling capacity of recurrent networks. Such hybrid structures have demonstrated strong performance in multivariate time-series forecasting settings Zhang et al. (2024) and are particularly suited for signals exhibiting both short-term fluctuations and longer periodic patterns.

Training Configuration

All deep learning models were implemented in Keras and trained using the Adam optimizer with mean squared error (MSE) as the primary loss function. Early stopping based on validation loss was applied to prevent overfit-

ting. Dropout regularization and batch normalization were incorporated where appropriate to enhance generalization stability.

The final output layer of each architecture produces a 24-dimensional forecast vector in a single forward pass, ensuring alignment with the multi-horizon decision-support objective.

3.2.3 Horizon-Aware Hybrid Model

During the comparative evaluation of baseline models, a consistent pattern emerged: no single forecasting model achieved uniformly superior performance across the entire 24-hour prediction horizon. As illustrated in Figures 4.1 and 4.3, certain models demonstrated strong accuracy for short-term horizons (e.g., 1–6 hours ahead), while other models exhibited lower error at medium- and long-term horizons (e.g., 12–24 hours ahead).

This horizon-dependent performance behavior indicates that selecting a single globally optimal model is inherently suboptimal for multi-step forecasting. A model that minimizes average error across all horizons may still perform poorly at specific decision-critical lead times. Since the objective of this thesis is to provide a reliable 24-hour availability trajectory for decision-support purposes, robustness across horizons is more important than dominance at a single aggregated metric.

To address this limitation, a hybrid forecast combination strategy is proposed. Rather than relying on one model exclusively, predictions from two complementary base models are combined. The central idea is that different models capture different structural properties of the data: some are more responsive to short-term fluctuations, while others better capture smoother long-range trends. By blending their forecasts, the hybrid approach aims to exploit complementary strengths and mitigate individual weaknesses.

Forecast combination is a well-established concept in time series forecasting and ensemble learning, where combining diverse estimators often leads to improved stability and generalization. However, traditional forecast blending typically applies a single global weight across all forecast horizons. In contrast, the empirical findings in this study suggest that the optimal combination weight may vary systematically with lead time.

Therefore, two variants of a horizon-aware blending strategy are investigated:

- **Global Weighted Blend (GWB):** a single shared blending coefficient is applied uniformly across all 24 forecast horizons.
- **Local Weighted Blend (LWB):** horizon-specific blending coefficients are estimated independently for each forecast step.

The Global Weighted Blend represents a parsimonious and easily deployable ensemble, while the Local Weighted Blend explicitly models horizon-dependent performance differences. This design allows systematic evaluation of whether allowing horizon-specific adaptation yields measurable improvements in predictive accuracy and robustness.

Selection of Complementary Base Models

Because parking availability dynamics differ between facilities in terms of capacity, sensor density, and demand variability, base model selection was conducted

independently for each parking lot while maintaining an identical hybrid formulation.

The objective of the selection procedure was to identify two models that (i) demonstrate strong overall predictive performance and (ii) exhibit complementary strengths across forecast horizons. Rather than selecting the two models with the lowest aggregated error alone, the procedure explicitly accounts for horizon-wise dominance.

The selection process consisted of four steps:

1. All candidate models were ranked according to validation RMSE aggregated across the full 24-hour horizon.
2. The top four models were retained to ensure that only competitively performing estimators were considered.
3. For each forecast horizon $h \in \{1, \dots, 24\}$, the best-performing model among the retained candidates was identified based on validation RMSE.
4. The frequency with which each model appeared as the horizon-wise winner was computed. The two models with the highest occurrence counts were selected as the complementary base pair (A, B) .

Formally, let $M = \{M_1, \dots, M_K\}$ denote the retained candidate models. For each horizon h , define

$$M^*(h) = \arg \min_{M_i \in M} \text{RMSE}_h(M_i),$$

and let

$$c_i = \sum_{h=1}^{24} \mathbf{1}(M^*(h) = M_i)$$

denote the number of horizons for which model M_i achieves the lowest validation error. The two models with the highest c_i values are selected as the hybrid base models.

This frequency-based selection criterion ensures that the chosen models are not only globally competitive but also consistently strong across multiple forecast horizons. By emphasizing horizon-wise dominance rather than aggregated performance alone, the procedure explicitly aligns base model selection with the multi-step forecasting objective.

Horizon-Aware Hybrid Forecast Formulation

Let (A, B) denote the complementary base model pair selected according to the procedure described in Section 3.2.3. For a given time index t and forecast horizon $h \in \{1, \dots, 24\}$, let $\hat{y}_{t,h}^{(A)}$ and $\hat{y}_{t,h}^{(B)}$ denote the predicted number of free parking spaces produced by models A and B , respectively.

The hybrid forecast is constructed as a convex combination of the two base predictions:

$$\hat{y}_{t,h}^{(H)} = \alpha_h \hat{y}_{t,h}^{(A)} + (1 - \alpha_h) \hat{y}_{t,h}^{(B)}, \quad \alpha_h \in [0, 1]. \quad (3.3)$$

The convexity constraint ensures that the hybrid prediction remains within the range spanned by the base model outputs, preserving stability and interpretability.

Two variants of this blending strategy are investigated: a Global Weighted Blend (GWB) and a Local Weighted Blend (LWB).

The selection of the base models (A, B) is guided by the empirical evaluation of baseline model performance presented in Chapter 4. For Lot 38, Linear Regression and the Encoder–Decoder model are chosen due to their complementary strengths across short- and long-term forecast horizons. For Lot 634, Gradient Boosting and Random Forest are selected based on their consistently strong performance across the evaluated horizons. This selection is not arbitrary but is grounded in the comparative analysis of baseline models, where the top-performing candidates are identified and their horizon-dependent behavior is examined. The results demonstrate that these model pairs exhibit complementary predictive characteristics, which motivates their use within the proposed hybrid framework.

Global Weighted Blend (GWB)

The Global Weighted Blend applies a single blending coefficient α across all forecast horizons:

$$\hat{y}_{t,h}^{(\text{GWB})} = \alpha \hat{y}_{t,h}^{(A)} + (1 - \alpha) \hat{y}_{t,h}^{(B)}, \quad \alpha \in [0, 1]. \quad (3.4)$$

The optimal blending coefficient α^* is determined using the validation set exclusively, thereby avoiding test data leakage. Let $y_{t,h}$ denote the observed validation targets. The optimal weight minimizes the aggregated validation RMSE across all samples and horizons:

$$\alpha^* = \arg \min_{\alpha \in [0,1]} \sqrt{\frac{1}{NH} \sum_{t=1}^N \sum_{h=1}^H \left(y_{t,h} - \hat{y}_{t,h}^{(\text{GWB})}(\alpha) \right)^2}, \quad H = 24. \quad (3.5)$$

In practice, α is selected via grid search over $\{0.00, 0.05, \dots, 1.00\}$. The resulting α^* is fixed and applied unchanged when generating test predictions.

The optimization of the global blending coefficient α is illustrated in Figures 3.11 and 3.12 for both parking facilities. Each curve represents the validation RMSE as a function of $\alpha \in [0, 1]$, reflecting the trade-off between the two base models.

For Lot 38, the optimal value is obtained at $\alpha \approx 0.45$, indicating a slightly higher contribution of the Encoder–Decoder model compared to Linear Regression. In contrast, for Lot 634, the optimal weight is $\alpha \approx 0.6$, assigning greater importance to the Gradient Boosting model relative to Random Forest.

These results confirm that the optimal blending weight is data-dependent and varies across parking facilities. The convex shape of the curves further demonstrates the benefit of combining complementary models, as intermediate α values consistently outperform the individual base models ($\alpha = 0$ or $\alpha = 1$).

3.2. MODEL DEVELOPMENT AND EVALUATION

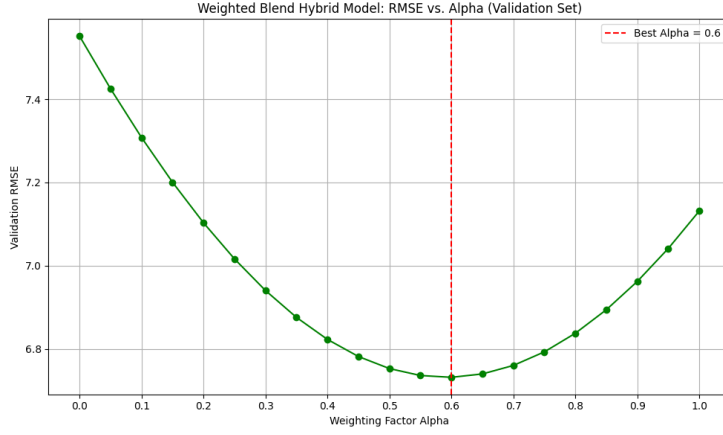


Figure 3.11: Validation RMSE vs. α for GWB (Lot 38).

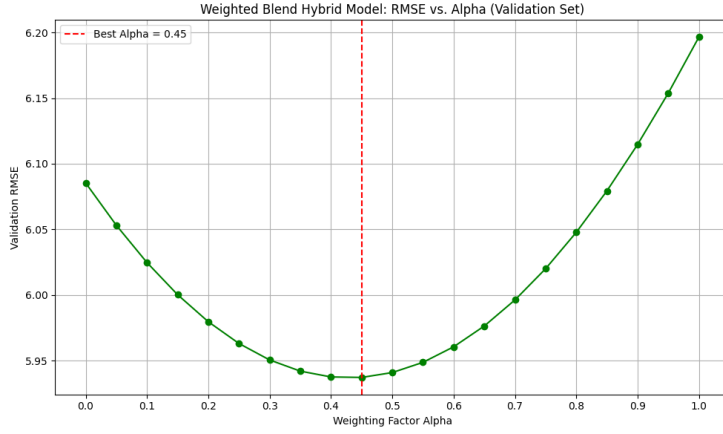


Figure 3.12: Validation RMSE vs. α for GWB (Lot 634).

Local Weighted Blend (LWB)

The Local Weighted Blend generalizes the global approach by allowing a distinct blending coefficient for each forecast horizon:

$$\hat{y}_{t,h}^{(\text{LWB})} = \alpha_h \hat{y}_{t,h}^{(A)} + (1 - \alpha_h) \hat{y}_{t,h}^{(B)}, \quad \alpha_h \in [0, 1]. \quad (3.6)$$

Each blending coefficient α_h is optimized independently using validation data:

$$\alpha_h^* = \arg \min_{\alpha_h \in [0,1]} \sqrt{\frac{1}{N} \sum_{t=1}^N \left(y_{t,h} - \hat{y}_{t,h}^{(\text{LWB})}(\alpha_h) \right)^2}. \quad (3.7)$$

Optimization is performed separately for each horizon via grid search over $\{0.00, 0.05, \dots, 1.00\}$. The resulting weight vector

$$\alpha^* = [\alpha_1^*, \dots, \alpha_{24}^*] \quad (3.8)$$

captures the empirically observed horizon-dependent dominance pattern identified during model comparison.

By allowing the blending coefficient to vary with lead time, the LWB formulation explicitly adapts to differences in short-, medium-, and long-term predictive strengths of the base models. This design directly aligns with the decision-support objective of delivering a stable and reliable 24-hour parking availability trajectory rather than optimizing only a globally averaged error metric.

The horizon-specific optimal blending coefficients obtained for both parking facilities are illustrated in Figures 3.13 and 3.14. The results show that the relative contribution of the two base models varies noticeably across forecast horizons, confirming that model dominance is not constant over time. For Lot 38, higher α_h values are observed for most short- and long-term horizons, indicating a stronger contribution of Linear Regression, while the weights become more balanced in the middle horizons. In contrast, for Lot 634, the optimized weights are generally lower for medium- and longer-term horizons, implying a stronger influence of Random Forest, whereas Gradient Boosting contributes more strongly in the earliest forecast steps. These patterns provide empirical support for the Local Weighted Blend formulation, showing that a horizon-specific weighting strategy can better reflect the changing predictive strengths of the base models than a single global coefficient.

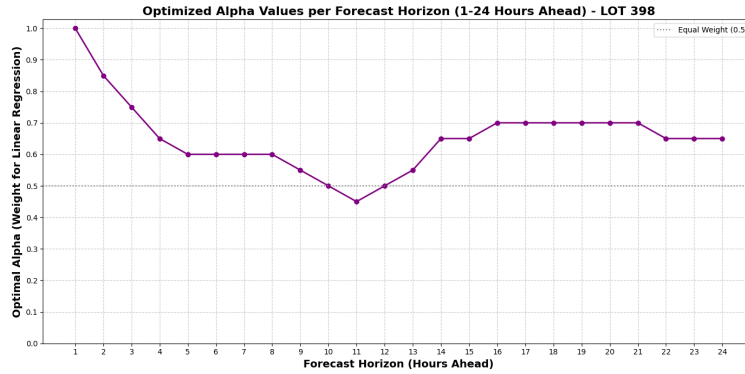


Figure 3.13: Horizon-specific optimal weights α_h for the LWB model (Lot 38).

3.2.4 Feature Importance Analysis

To identify the most influential predictors for parking availability, feature importance was analyzed using a Random Forest regression model.

Random Forest is an ensemble learning method composed of multiple decision trees, each trained on a bootstrap sample of the data and using random subsets of features for splitting. This design reduces overfitting and enables the model to capture nonlinear relationships and interactions between variables.

In regression tasks, decision trees evaluate the quality of a split based on the reduction in variance of the target variable. Let Y denote the target variable (number of free parking spaces). For a node containing N samples, the variance is defined as:

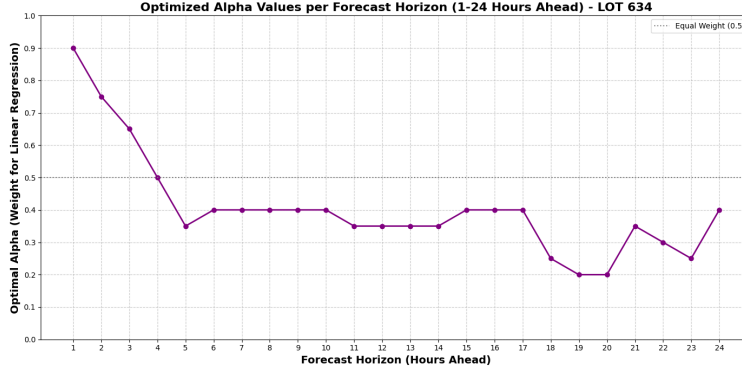


Figure 3.14: Horizon-specific optimal weights α_h for the LWB model (Lot 634).

$$\text{Var}(Y) = \frac{1}{N} \sum_{i=1}^N (y_i - \bar{y})^2 \quad (3.9)$$

When splitting a node using a feature X_j , the data is divided into left and right child nodes. The quality of the split is measured by the reduction in variance:

$$\Delta \text{Var} = \text{Var}_{\text{parent}} - \left(\frac{N_{\text{left}}}{N} \text{Var}_{\text{left}} + \frac{N_{\text{right}}}{N} \text{Var}_{\text{right}} \right) \quad (3.10)$$

This quantity reflects how much the split reduces prediction uncertainty. Features that produce larger reductions are considered more informative.

During training, each tree selects splits that maximize this variance reduction. The Random Forest aggregates these contributions across all trees. The importance of a feature X_j is computed as the total variance reduction it contributes across all splits and trees:

$$\text{Importance}(X_j) = \sum_{t=1}^T \sum_{n \in \text{splits}(X_j)} \Delta \text{Var}_{n,t} \quad (3.11)$$

These values are normalized such that the total importance across all features sums to one.

Since the input data includes multiple time-lagged versions of each variable, each base feature appears several times in the feature vector. To improve interpretability, importance values were aggregated by summing contributions across all lagged instances of the same base feature. The resulting values represent the overall influence of each predictor on the forecasting task.

3.2.5 Implementation Environment

All experiments were conducted using Python 3 within the Google Colab environment. Classical machine learning models were implemented using the scikit-learn library, while deep learning architectures were developed using TensorFlow and Keras.

Training was performed on GPU-accelerated hardware (NVIDIA A100) to ensure computational efficiency, particularly for recurrent and hybrid deep learning architectures. The same runtime configuration was used consistently across all experiments to ensure comparability of results.

Data preprocessing, feature engineering, and model evaluation were integrated into a unified training pipeline to guarantee reproducibility and consistent experimental conditions.

3.2.6 Evaluation Metrics

The predictive performance of all models was assessed using standard regression metrics tailored to the multi-step forecasting setting. Since the objective is to generate a full 24-hour availability trajectory, evaluation must reflect accuracy across the entire forecast horizon rather than at a single step.

Let N denote the number of test samples and H the prediction horizon (with $H = 24$). For each sample $i \in \{1, \dots, N\}$ and horizon step $h \in \{1, \dots, H\}$, let $y_{i,h}$ denote the observed number of free parking spaces and $\hat{y}_{i,h}$ the corresponding model prediction. All metrics are computed over the complete set of $N \times H$ predictions to ensure a consistent and comparable assessment across models.

Mean Squared Error (MSE)

The Mean Squared Error penalizes large deviations quadratically and therefore emphasizes substantial forecasting errors. It is widely used both as an evaluation metric and as a training objective for neural forecasting models:

$$\text{MSE} = \frac{1}{NH} \sum_{i=1}^N \sum_{h=1}^H (y_{i,h} - \hat{y}_{i,h})^2. \quad (3.12)$$

Root Mean Squared Error (RMSE)

The Root Mean Squared Error is obtained by taking the square root of MSE and therefore shares the same unit as the target variable (number of free parking spaces), which facilitates interpretability in operational terms:

$$\text{RMSE} = \sqrt{\text{MSE}}. \quad (3.13)$$

Mean Absolute Error (MAE)

The Mean Absolute Error measures the average absolute deviation between predictions and observations. Compared to RMSE, it is less sensitive to outliers and therefore provides a complementary robustness-oriented performance perspective:

$$\text{MAE} = \frac{1}{NH} \sum_{i=1}^N \sum_{h=1}^H |y_{i,h} - \hat{y}_{i,h}|. \quad (3.14)$$

Coefficient of Determination (R^2)

The coefficient of determination quantifies the proportion of variance explained by the model relative to a mean predictor baseline. Let $\bar{y}$ denote the mean of all ground-truth values across samples and horizons:

$$\bar{y} = \frac{1}{NH} \sum_{i=1}^N \sum_{h=1}^H y_{i,h}. \quad (3.15)$$

The R^2 score is defined as:

$$R^2 = 1 - \frac{\sum_{i=1}^N \sum_{h=1}^H (y_{i,h} - \hat{y}_{i,h})^2}{\sum_{i=1}^N \sum_{h=1}^H (y_{i,h} - \bar{y})^2}. \quad (3.16)$$

Values closer to 1 indicate stronger explanatory power, whereas negative values indicate performance worse than a naive mean predictor.

3.3 Deployment and System Architecture

To operationalize the proposed forecasting framework in a real-world setting, the complete end-to-end pipeline was deployed on an Amazon Web Services (AWS) EC2 instance using a containerized, service-oriented architecture. The deployment integrates data ingestion, preprocessing, forecasting, storage, and visualization into a unified real-time decision-support system.

Figure 3.15 illustrates the overall system architecture and data flow. External data sources include a Parking Availability API and a Weather API, which provide real-time observations. These external streams are continuously ingested by a dedicated data collection service running within the EC2 environment. Raw observations are persisted without modification in a centralized PostgreSQL database to ensure traceability and reproducibility.

A transformation service periodically performs SQL-based aggregation and feature preparation directly within the database layer. This includes converting event-level parking data into hourly availability statistics and aligning them with corresponding weather observations and calendar descriptors, resulting in model-ready structured inputs.

The forecasting service loads the trained Local Weighted Blend (LWB) hybrid model and generates a 24-hour ahead availability trajectory for each parking facility. Predicted values are written back into PostgreSQL in dedicated forecast tables, separating operational outputs from historical observations.

Finally, a lightweight web dashboard service queries the latest prediction tables and presents the next 24-hour availability forecast to end users via an interactive interface. This completes the real-time inference loop from data acquisition to user-facing decision support.

All internal components (data collection, transformation, forecasting, dashboard, and database) are deployed as isolated Docker containers and orchestrated using Docker Compose. This modular design ensures reproducibility, clear separation of responsibilities, simplified dependency management, and consistent startup behavior across environments. The architecture supports maintainability and extensibility while remaining computationally efficient for deployment on a single cloud instance.

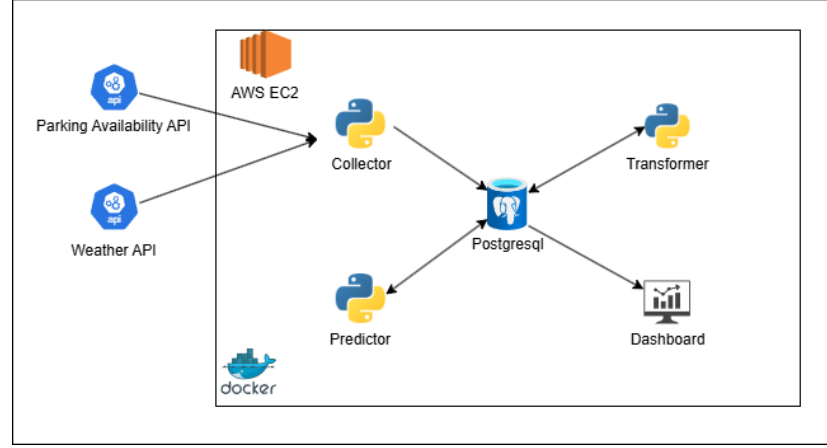


Figure 3.15: Containerized cloud deployment architecture.

3.3.1 Data Collection Component

The data collection service is responsible for continuously retrieving real-time operational and environmental data and persisting them in the central PostgreSQL database. The service is implemented in Python and deployed as isolated Docker containers within the AWS EC2 environment (cf. Figure 3.15). Its primary function is to provide a consistent and up-to-date raw data layer for subsequent transformation, feature construction, and forecasting stages.

Parking Availability Data

Real-time parking availability observations are obtained from the external Parking Availability API. Authentication is handled via a token-based mechanism, where an access token is requested using API credentials and automatically refreshed upon expiration to ensure uninterrupted operation. The endpoint providing the current occupancy state of all parking facilities is queried at a fixed interval of one minute.

Each response is parsed into a structured format containing a timestamp, parking lot identifier, and occupancy indicators. The forecasting target of this thesis, the number of free parking spaces, is derived from total capacity and occupied spaces. The processed observations are written into a dedicated relational table (*parking_availability_minute*) with minute-level granularity. The schema includes a timestamp, parking lot identifier, availability indicators (free and occupied spaces, total capacity, occupancy rate), and basic facility meta-data.

Weather Data

Hourly meteorological observations are retrieved from the Open-Meteo API for the geographic coordinates of the study area (Bamberg, Germany). The service collects key environmental variables, including temperature at 2 meters, relative humidity, precipitation, wind speed, and weather condition codes. Weather data

are fetched once per hour to align with the temporal resolution of the forecasting task.

Observations are stored in a dedicated relational table (*weather_hourly*) at hourly resolution. The schema includes an hourly timestamp (*hour_timestamp*) and the selected meteorological indicators. To ensure idempotent operation and avoid duplicate records, the table enforces a uniqueness constraint on the timestamp, and insert operations apply an upsert strategy when necessary.

Together, the minute-level parking table and the hourly weather table form the raw operational data layer of the system. These tables serve as the input for the subsequent SQL-based transformation and integration step that produces model-ready hourly datasets.

3.3.2 Data Transformation Component

The data transformation component converts raw minute-level parking observations and hourly weather data into model-ready features for multi-step forecasting. It operates as an intermediate processing layer between data collection and model inference and is deployed as a dedicated Docker container within the AWS EC2 environment.

Hourly Aggregation

The first stage aggregates minute-level parking availability data into hourly summaries. For each parking facility, statistical measures such as average, minimum, and maximum free spaces are computed over hourly windows. This transformation reduces high-frequency noise while preserving the temporal structure required for forecasting. Aggregation is performed consistently across all facilities to ensure temporal alignment with hourly weather observations.

Feature Engineering and Integration

In the second stage, aggregated parking availability is joined with the corresponding hourly weather data stored in PostgreSQL. The unified dataset forms the basis for feature construction.

Feature engineering includes:

- Calendar-based descriptors (e.g., hour of day, day of week, month),
- Rolling statistics capturing short-term dynamics (e.g., 24-hour rolling mean and standard deviation),
- First-order differences to represent temporal changes,
- Derived contextual indicators based on weather conditions.

The resulting hourly feature table represents the final input structure for supervised learning and hybrid forecasting.

Scheduling and Orchestration

The transformation workflow is triggered automatically on an hourly schedule after new weather and parking data have been collected. Execution is coordinated through a centralized scheduler to ensure correct processing order and

temporal consistency. All transformations are performed directly within the PostgreSQL database using SQL-based operations to guarantee efficiency and transactional integrity.

This component establishes the structured feature layer required for reliable next-day parking availability prediction while maintaining compatibility with the real-time deployment pipeline.

3.3.3 Prediction Component

The prediction component represents the analytical core of the deployed system. It generates 24-hour ahead forecasts of parking availability using the proposed Local Weighted Blend Hybrid model. The service operates as an independent Docker container within the AWS EC2 environment and is executed on an hourly schedule after completion of the data transformation stage.

At each prediction cycle, the service retrieves the most recent 48 hours of aggregated parking availability and integrated weather features from PostgreSQL. These inputs are processed using the same preprocessing configuration applied during model training (including scaling and feature alignment) to ensure consistency between training and deployment environments. The Local Hybrid model then produces a multi-step forecast covering the next 24 hours at hourly resolution.

Separate trained hybrid models are maintained for each parking facility, reflecting location-specific demand dynamics. Model artifacts (including base models and blending weights) are loaded dynamically at runtime, allowing independent updates without modifying the overall pipeline structure.

The generated 24-hour forecasts are stored in a dedicated PostgreSQL table designed for time-series prediction outputs. This table serves as the data source for the web-based dashboard and enables real-time visualization of expected parking availability. The prediction workflow is fully automated and coordinated through a scheduled execution mechanism to ensure temporal consistency and continuous operation.

3.3.4 Interactive Dashboard Component

The dashboard component provides a web-based interface for visualizing real-time parking availability and 24-hour forecasts generated by the Local Hybrid prediction model. It is implemented using the Flask framework and deployed as a Docker container within the AWS EC2 environment. The dashboard serves as the final presentation layer of the system, enabling users to access forecast results through a browser-based application.

The application retrieves predicted availability data directly from the PostgreSQL database. For each parking facility, the dashboard displays hourly forecasts for the next 24 hours using interactive time-series visualizations. Availability levels are represented through clear graphical indicators and color-coded elements to support rapid interpretation of expected parking capacity.

To ensure up-to-date information, the dashboard queries the prediction table at regular intervals and reflects the most recent forecast cycle. The architecture follows a server-side rendering approach combined with lightweight client-side interactions, balancing performance and responsiveness.

3.3. DEPLOYMENT AND SYSTEM ARCHITECTURE

Overall, the dashboard acts as the user-facing layer of the deployed forecasting pipeline, translating machine learning predictions into intuitive visual outputs suitable for operational decision support.

Chapter 4

Results and Evaluation

This chapter presents the empirical evaluation of the proposed parking availability forecasting framework. The objective is to systematically compare the predictive performance of classical machine learning models, deep learning architectures, and the proposed hybrid forecast combination methods under a consistent experimental protocol.

We begin by analyzing the performance of all baseline models introduced in 3. Classical regression models and deep sequence architectures are evaluated using identical train-validation-test splits and the multi-step forecasting setup described previously. Based on aggregated accuracy metrics, the top-performing models are identified for each parking facility.

Next, we conduct a horizon-wise performance analysis to investigate how prediction accuracy varies across the 24-hour forecast horizon. This analysis highlights that no single model consistently achieves the lowest error at all forecast steps. Instead, different models exhibit complementary strengths at short- and long-term horizons. The four strongest baseline models are therefore examined in greater detail to identify horizon-specific winners for each parking facility.

Building upon these observations, the results of the Global Weighted Blend (GWB) and Local Weighted Blend (LWB) hybrid approaches are presented. Their predictive performance is compared against the best individual baseline models using standard regression metrics. Special attention is given to evaluating whether horizon-aware blending provides measurable improvements over single-model forecasting strategies.

In addition to predictive accuracy, feature importance analysis is performed for the most relevant models to quantify the contribution of calendar, weather, and temporal dynamics features. This analysis provides insights into the drivers of parking availability and supports interpretability of the forecasting framework.

Finally, we present the operational results of the deployed dashboard system. Example visualizations illustrate how 24-hour forecasts are presented to end users, demonstrating the practical integration of the prediction pipeline into a real-time decision-support interface.

The following sections report quantitative findings without interpretation; implications and theoretical insights are discussed in Chapter 3.

4.1 Overall Baseline Models Performance

This section compares the predictive performance of all baseline models for each parking facility separately. Model ranking is primarily based on RMSE aggregated across the full 24-hour forecasting horizon, while MAE and R^2 are reported for completeness. Based on these results, the top four performing models for each parking lot are identified and selected for further analysis and inclusion in the hybrid forecasting framework.

Lot 38

Table 4.1 summarizes the performance results for Lot 38. Among all evaluated models, the Encoder–Decoder architecture achieved the lowest RMSE (11.496), followed by Linear Regression (12.226), Ridge Regression (12.344), and LSTM (12.562). These four models therefore constitute the top-performing candidates for this facility.

The results indicate that both deep learning and linear classical models perform competitively for Lot 38. In contrast, tree-based ensemble methods such as Gradient Boosting, Random Forest, and XGBoost exhibit substantially higher errors, with negative R^2 values, indicating performance below the mean baseline predictor.

Based on aggregated RMSE, the top four models retained for detailed horizon-wise analysis in Lot 38 are:

- Encoder–Decoder
- Linear Regression
- Ridge Regression
- LSTM

Table 4.1: Baseline model performance for Lot 38 (24-hour forecast).

Model	RMSE	MAE	R^2
Encoder–Decoder	11.496	9.566	0.666
Linear Regression	12.226	8.567	0.620
Ridge Regression	12.344	8.659	0.613
LSTM	12.562	10.734	0.601
BiLSTM	13.084	10.867	0.567
CNN–LSTM	14.096	12.103	0.497
GRU	14.420	12.235	0.474
CNN–GRU	16.351	13.802	0.323
Gradient Boosting	20.480	16.243	-0.064
Random Forest	20.692	16.002	-0.085
XGBoost	20.733	16.267	-0.091
Decision Tree	20.983	16.548	-0.117
SVR	27.349	24.311	-0.893

Lot 634

The performance results for Lot 634 are presented in Table 4.2. In contrast to Lot 38, tree-based ensemble methods dominate the ranking. Gradient Boosting achieves the lowest RMSE (5.217), closely followed by Random Forest (5.236) and XGBoost (5.290). Ridge Regression (5.724) completes the top four.

All models achieve substantially lower absolute error values compared to Lot 38, and most methods obtain high R^2 values above 0.85, indicating strong explanatory power.

For Lot 634, the four best-performing baseline models retained for further horizon-wise investigation are:

- Gradient Boosting
- Random Forest
- XGBoost
- Ridge Regression

In the next step, we analyze the horizon-dependent behavior of these top-performing models to determine whether a single model consistently dominates across all forecast steps.

Table 4.2: Baseline model performance for Lot 634 (24-hour forecast).

Model	RMSE	MAE	R^2
Gradient Boosting	5.217	3.374	0.914
Random Forest	5.236	3.432	0.914
XGBoost	5.290	3.413	0.912
Ridge Regression	5.724	3.792	0.897
Linear Regression	5.777	3.854	0.895
BiLSTM	5.977	4.112	0.888
Encoder–Decoder	6.072	4.025	0.884
GRU	6.254	4.434	0.877
LSTM	6.481	4.635	0.868
SVR	6.500	4.511	0.867
CNN–GRU	6.654	4.835	0.861
CNN–LSTM	6.833	4.842	0.853
Decision Tree	7.477	4.840	0.824

4.2 Performance Across Forecast Horizons

This section analyzes model performance across different forecast horizons to evaluate how predictive accuracy evolves over time. Rather than relying solely on aggregated error metrics, performance is examined at each prediction step within the 24-hour horizon, enabling the identification of horizon-dependent behavior. This analysis highlights how different models perform across short-, medium-, and long-term forecasts and reveals complementary strengths among

model families. These findings are subsequently used to determine the components of the proposed hybrid approaches, namely the Global Weighted Blend (GWB) and Local Weighted Blend (LWB), which leverage horizon-specific performance characteristics.

Lot 38

Figure 4.1 presents the RMSE for each forecast horizon ($h = 1, \dots, 24$) for the four strongest baseline models on Lot 38. The results clearly show that model performance is not uniform across the forecasting horizon.

For short forecast horizons (approximately $h = 1$ to $h = 8$), Linear Regression and Ridge Regression achieve the lowest RMSE values among the four candidates. In particular, Linear Regression consistently attains the minimum error in the earliest steps, indicating superior short-term accuracy compared to the deep learning models.

From approximately $h = 9$ onward, the performance ranking changes. The Encoder–Decoder model increasingly outperforms the linear models at longer horizons, achieving lower RMSE values in the majority of forecast steps between $h = 13$ and $h = 24$. LSTM also shows competitive performance at intermediate and longer horizons but does not dominate as consistently as the Encoder–Decoder architecture.

In contrast, the Encoder–Decoder and LSTM models exhibit higher errors in the first forecast steps compared to the linear models, indicating weaker short-term performance. Meanwhile, Linear Regression and Ridge Regression show a steady increase in error as the forecast horizon grows, becoming less competitive at longer lead times.

Table 4.2 summarizes the horizon-wise winners among the top four models. Counting the number of horizons in which each model achieves the lowest RMSE shows that Linear Regression dominates short horizons, whereas the Encoder–Decoder dominates long horizons.

Based on this empirical distribution of horizon-wise winners, the two base models selected for the proposed hybrid approach on Lot 38 are:

- Linear Regression (strongest in short horizons)
- Encoder–Decoder (strongest in long horizons)

This selection directly reflects the observed complementary performance across the 24-hour forecast horizon.

Lot 634

Figure 4.3 presents the RMSE across the 24-hour forecast horizon for the four best-performing baseline models on Lot 634: Gradient Boosting, Random Forest, XGBoost, and Ridge Regression.

In contrast to Lot 38, performance differences among the top models are relatively small, particularly in short forecast horizons. For the first hours ($h = 1$ to approximately $h = 8$), competition is close between Gradient Boosting, XGBoost, and Ridge Regression, while Random Forest generally exhibits slightly higher errors in this early range. Among these models, Gradient Boosting most frequently achieves the lowest RMSE in short horizons.

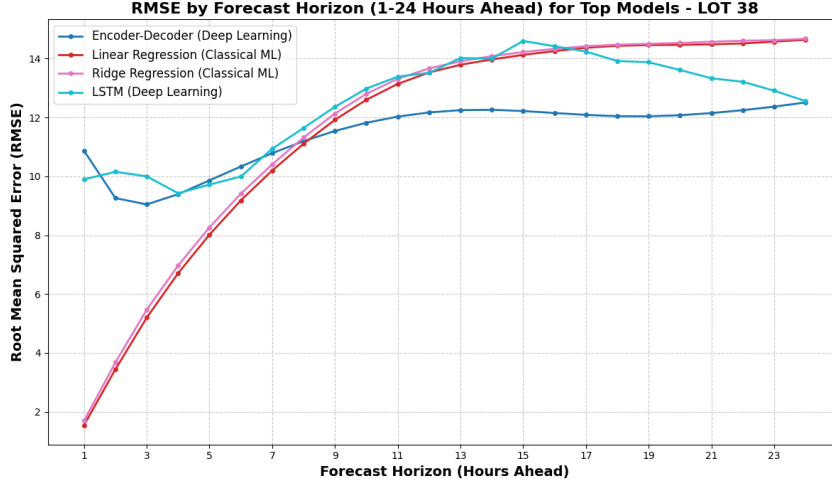


Figure 4.1: RMSE by forecast horizon for the top four baseline models on Lot 38.

As the forecast horizon increases, the ranking shifts. From intermediate to longer horizons (approximately $h \geq 12$), Random Forest increasingly achieves lower RMSE values compared to the other models. In the final forecast steps, Random Forest becomes the dominant performer, while Ridge Regression consistently exhibits the highest error among the top four models in longer horizons.

Table 4.4 summarizes the horizon-wise winners across all 24 forecast steps. Counting the number of horizons in which each model achieves the minimum RMSE shows that Gradient Boosting dominates short horizons, whereas Random Forest dominates longer horizons.

Based on this empirical distribution of horizon-wise winners, the two base models selected for the proposed hybrid approach on Lot 634 are:

- Gradient Boosting (strongest in short horizons)
- Random Forest (strongest in long horizons)

This selection reflects the observed complementary performance patterns across the forecast horizon.

4.3 Hybrid Model vs Baselines

This section compares the performance of the proposed horizon-aware hybrid models against the selected baseline models. The evaluation focuses on both overall predictive accuracy and horizon-specific performance across the 24-hour forecasting window. By benchmarking the Global Weighted Blend (GWB) and Local Weighted Blend (LWB) against the top-performing individual models, the analysis assesses whether combining models based on their complementary strengths leads to measurable improvements. The results provide insight into the effectiveness of the hybrid approach in addressing horizon-dependent variability observed in the baseline models.

Forecast Horizon (Hours)	Encoder-Decoder	Linear Regression	Ridge Regression	LSTM
1	10.870017	1.543910	1.692180	9.897806
2	9.260547	3.442075	3.676166	10.153080
3	9.044999	5.198048	5.469697	9.999928
4	9.394036	6.707784	6.969137	9.423997
5	9.862511	8.019120	8.265733	9.719165
6	10.334182	9.184230	9.411513	9.995040
7	10.787165	10.200530	10.407828	10.938784
8	11.194653	11.108086	11.319516	11.645024
9	11.541838	11.921861	12.134547	12.363406
10	11.820171	12.600015	12.803349	12.981030
11	12.029202	13.140809	13.313800	13.386150
12	12.171275	13.528221	13.668002	13.516698
13	12.248671	13.791369	13.910249	14.018999
14	12.260947	13.972317	14.083594	14.000582
15	12.218507	14.127366	14.222808	14.599591
16	12.151795	14.254987	14.332159	14.419956
17	12.087769	14.369094	14.420944	14.239099
18	12.044033	14.434809	14.473388	13.919716
19	12.040311	14.460324	14.502599	13.877016
20	12.077156	14.467251	14.533417	13.613220
21	12.148683	14.484410	14.573344	13.331997
22	12.247703	14.516981	14.609224	13.211413
23	12.367200	14.575177	14.624831	12.910361
24	12.509980	14.639322	14.674674	12.556189

Figure 4.2: Horizon-wise best model for Lot 38 (lowest RMSE).

Lot 38

Table 4.3 summarizes the overall multi-step forecasting accuracy over the full 24-hour horizon for Lot 38. Both proposed hybrid variants outperform all standalone baseline models. The Local Weighted Blend Hybrid achieves the best overall performance (RMSE = 10.631, MAE = 8.023, $R^2 = 0.713$), followed closely by the Global Weighted Blend Hybrid (RMSE = 10.665, MAE = 8.090, $R^2 = 0.711$). In comparison, the best single model, the Encoder-Decoder, reaches an RMSE of 11.496.

To assess horizon-wise behavior, Figure 4.5 compares the Local Weighted Blend Hybrid to its two component models (Linear Regression and Encoder-Decoder) across forecast horizons. The results show that Linear Regression performs best at the very short horizons, while the Encoder-Decoder is consistently stronger at longer lead times. The Local Weighted Blend Hybrid matches the strongest model at short horizons and yields lower errors for almost all medium and long horizons. Apart from a small difference at a few very early steps, the hybrid achieves the lowest RMSE across the majority of the 24-hour horizon, confirming that horizon-aware blending improves accuracy relative to selecting a single model for all lead times.

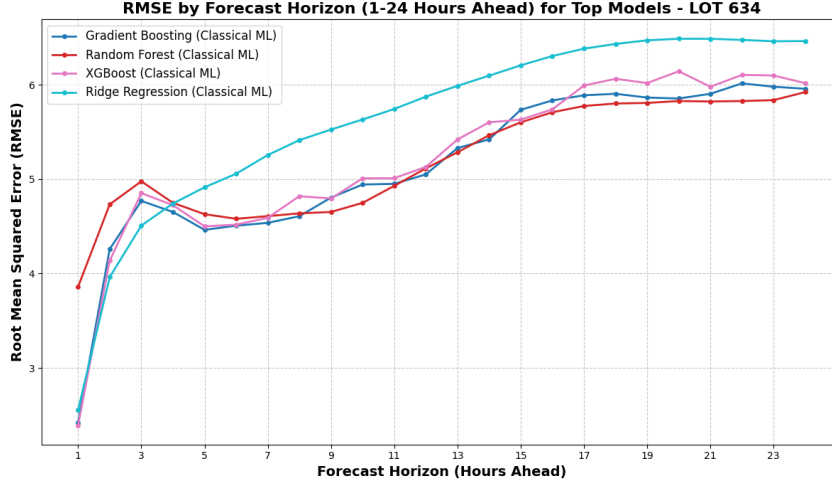


Figure 4.3: RMSE by forecast horizon for the top four baseline models on Lot 634.

Lot 634

Table 4.4 presents the aggregated multi-step forecasting performance for Lot 634. Both proposed hybrid approaches achieve the best overall accuracy among all evaluated models. The Local Weighted Blend Hybrid attains the lowest RMSE (5.060), lowest MAE (3.265), and highest R^2 (0.920). The Global Weighted Blend Hybrid follows closely with RMSE = 5.067 and R^2 = 0.919.

Compared to the strongest standalone baseline model (Gradient Boosting, RMSE = 5.217), both hybrid variants provide a measurable improvement in overall accuracy.

Figure 4.6 compares the Local Weighted Blend Hybrid to its two component models, Gradient Boosting and Random Forest, across the 24 forecast horizons. The results show that the Local Hybrid achieves lower RMSE values for nearly all forecast steps. In the first forecast hour, the difference is minimal, but from hour 2 onward the hybrid consistently matches or outperforms both base models.

Across medium and longer horizons, the Local Hybrid maintains the lowest RMSE values with stable performance improvements relative to both Gradient Boosting and Random Forest. This confirms that combining the two complementary models results in superior accuracy across the full 24-hour trajectory rather than only in specific segments of the forecast horizon.

In summary, the experimental results confirm that the proposed hybrid forecasting approach provides the most accurate predictions for both parking facilities. The Local and Global Weighted Blend Hybrid models achieve the best overall performance across the full 24-hour prediction horizon while also maintaining strong accuracy at individual forecast steps. The horizon-wise analysis shows that no single baseline model consistently performs best for all horizons, whereas the hybrid models successfully capture the strengths of multiple base models. Consequently, the hybrid approach delivers improved and more stable forecasting accuracy across both short-term and long-term prediction horizons for both parking lots.

	Gradient Boosting	Random Forest	XGBoost	Ridge Regression
Forecast Horizon (Hours)				
1	2.418699	3.859352	2.392640	2.554331
2	4.258441	4.732690	4.136423	3.961770
3	4.769260	4.977181	4.852217	4.507989
4	4.652744	4.748933	4.724705	4.742048
5	4.462973	4.629117	4.500030	4.913936
6	4.507241	4.580246	4.517869	5.057695
7	4.537177	4.608301	4.590663	5.256002
8	4.607835	4.637771	4.819880	5.413863
9	4.804657	4.652459	4.795228	5.526170
10	4.943501	4.749153	5.009524	5.632298
11	4.950442	4.929142	5.010346	5.743592
12	5.050778	5.114676	5.129892	5.874620
13	5.330010	5.284986	5.420664	5.988956
14	5.421417	5.465168	5.602587	6.096748
15	5.734634	5.602319	5.631094	6.206430
16	5.834413	5.709079	5.739712	6.304640
17	5.888829	5.775439	5.992063	6.383104
18	5.904970	5.802037	6.063865	6.433046
19	5.864750	5.807799	6.018486	6.471677
20	5.854859	5.828421	6.143379	6.488139
21	5.904845	5.823541	5.979560	6.487598
22	6.015100	5.828148	6.105035	6.475889
23	5.979995	5.837716	6.098679	6.461462
24	5.957885	5.922857	6.017418	6.463439

Figure 4.4: Horizon-wise best model for Lot 634 (lowest RMSE).

4.3.1 Feature Importance Analysis

To identify the most influential predictors for parking availability forecasting, feature importance values were computed using the Random Forest model described in Section 3.2.4. The importance scores were aggregated across temporal lags to obtain the overall contribution of each base feature.

Lot 38

Figure 4.7 presents the top fifteen most important base features for Lot 38. Similar to the results observed for Lot 634, temporal variables play a dominant role in the prediction task. The feature *hour_of_day* clearly emerges as the most influential predictor, accounting for the largest portion of the total importance. This indicates that parking availability follows strong daily usage patterns.

Among the remaining predictors, several variables describing recent parking availability appear prominently. Features such as *avg_free_spaces*, *min_free_spaces*, and *avg_free_spaces_rolling_24h* provide information about recent occupancy conditions and therefore contribute significantly to the prediction performance. The variable *day_of_week* also ranks among the most relevant predictors, reflecting differences in parking demand across weekdays.

Weather-related variables, including *temperature_2m*, *temperature_diff_1h*, and *relative_humidity_2m*, appear with relatively smaller importance values compared to temporal and occupancy-related features. Calendar-based variables

Table 4.3: Overall performance of all evaluated models for Lot 38.

Model	RMSE	MAE	R ²
Local Weighted Blend Hybrid	10.631	8.023	0.713
Global Weighted Blend Hybrid	10.665	8.090	0.711
Encoder–Decoder	11.496	9.566	0.666
Linear Regression	12.226	8.567	0.620
Ridge Regression	12.344	8.659	0.613
LSTM	12.562	10.734	0.601
BiLSTM	13.084	10.868	0.567
CNN–LSTM	14.096	12.103	0.497
GRU	14.420	12.235	0.474
CNN–GRU	16.351	13.802	0.323
Gradient Boosting	20.480	16.243	-0.064
Random Forest	20.692	16.002	-0.085
XGBoost	20.733	16.267	-0.091
Decision Tree	20.983	16.548	-0.117
SVR	27.343	24.311	-0.893

such as *calendar_week*, *day_type*, and *day_of_month* show the lowest importance in the model.

Overall, the results for Lot 38 confirm that temporal patterns and recent parking availability statistics are the most important drivers of parking availability prediction, while weather and broader calendar features contribute comparatively less to the model’s predictive performance.

Lot 634

Figure 4.8 shows the top fifteen most important base features for Lot 634. The results indicate that temporal variables dominate the prediction task. In particular, *hour_of_day* is by far the most influential feature, accounting for the largest share of the total importance. This suggests that parking availability follows a strong daily cycle. The second most influential feature is *day_of_week*, indicating that parking demand differs across weekdays.

In addition to temporal features, several variables describing recent parking availability appear among the most important predictors. These include *avg_free_spaces*, *avg_free_spaces_diff_1h*, and *min_free_spaces*. These features capture short-term occupancy dynamics and provide the model with information about recent parking utilization patterns. Rolling statistics such as *avg_free_spaces_rolling_24h* and *avg_free_spaces_std_24h* also contribute to the prediction by representing longer-term temporal behavior.

Weather-related variables such as *temperature_2m*, *temperature_diff_1h*, *relative_humidity_2m*, and *precipitation* appear with smaller importance values compared to temporal and occupancy-based features. Calendar-based variables such as *calendar_week* and *day_of_month* also show relatively low importance in the prediction model.

Overall, the feature importance results indicate that parking availability prediction is primarily driven by temporal patterns and recent occupancy in-

	Linear Regression	Encoder Decoder	Local Weighted Blend Hybrid
Forecast Horizon (Hours)			
1	1.54	19.18	1.54
2	3.44	15.15	4.34
3	5.20	14.18	6.32
4	6.71	13.85	7.93
5	8.02	13.74	8.97
6	9.18	13.76	9.64
7	10.20	13.83	10.27
8	11.11	13.92	10.84
9	11.92	13.97	11.44
10	12.60	13.99	11.94
11	13.14	14.05	12.36
12	13.53	14.16	12.60
13	13.79	14.30	12.84
14	13.97	14.47	13.09
15	14.13	14.65	13.29
16	14.25	14.84	13.52
17	14.37	15.04	13.70
18	14.43	15.23	13.85
19	14.46	15.43	13.98
20	14.47	15.64	14.10
21	14.48	15.84	14.23
22	14.52	16.03	14.37
23	14.58	16.19	14.49
24	14.64	16.37	14.57

Figure 4.5: Horizon-wise best model for Lot 38.

formation, while weather and broader calendar variables play a comparatively smaller role in determining parking availability for Lot 634.

4.4 Deployment Results: Interactive Forecast Dashboard

To demonstrate the practical applicability of the proposed forecasting pipeline, the prediction system was deployed as a web-based dashboard that visualizes real-time parking availability and 24-hour forecasts for the selected parking facilities in Bamberg (see Figure 4.9). The dashboard serves as the user-facing interface of the system and allows users to access the forecasting results generated by the machine learning pipeline in an intuitive and interactive manner.

The dashboard is implemented as a web application and is deployed within the cloud-based infrastructure described in the system architecture. It is publicly accessible through the deployed API endpoint and can be opened in a standard web browser without requiring additional software. The current deployment can be accessed via the following address: <http://13.60.234.87:5000>

The platform integrates the outputs of the forecasting models with real-time parking data and relevant contextual information. The interface presents the predicted parking availability for the next 24 hours together with recent historical observations, enabling users to quickly assess expected parking conditions. By providing short-term forecasts of available parking spaces, the dashboard

4.4. DEPLOYMENT RESULTS: INTERACTIVE FORECAST DASHBOARD

Table 4.4: Overall performance of all evaluated models for Lot 634.

Model	RMSE	MAE	R ²
Local Weighted Blend Hybrid	5.060	3.265	0.920
Global Weighted Blend Hybrid	5.067	3.289	0.919
Gradient Boosting	5.217	3.374	0.914
Random Forest	5.236	3.432	0.914
XGBoost	5.290	3.413	0.912
Ridge Regression	5.724	3.792	0.897
Linear Regression	5.777	3.854	0.895
BiLSTM	5.977	4.112	0.888
Encoder–Decoder	6.072	4.025	0.884
GRU	6.254	4.434	0.877
LSTM	6.481	4.635	0.868
SVR	6.500	4.511	0.867
CNN–GRU	6.654	4.835	0.861
CNN–LSTM	6.833	4.842	0.853
Decision Tree	7.477	4.840	0.824

supports users in planning their arrival time and selecting an appropriate parking facility in advance.

To ensure that the displayed information reflects the most recent data, the dashboard updates automatically as new predictions are generated by the forecasting pipeline. The backend services continuously retrieve parking and weather data, update the forecasting models, and refresh the visualization layer accordingly. As a result, the dashboard provides a near real-time view of both current and predicted parking availability.

The interface consists of several interactive visual components designed to present the forecasting results in a clear and accessible format. These visualizations combine historical observations, future predictions, and contextual information within a unified interface. The individual charts and their functionality are discussed in detail in the following subsections.

4.4.1 Predicted Parking Availability Visualization

To present the forecasting results in an intuitive format, the dashboard visualizes the predicted parking availability for the next 24 hours using stacked bar charts (see Figure 4.10). Separate charts are provided for each monitored parking facility (Parking Lot 38 and Parking Lot 634). Each bar represents the predicted parking situation for a specific hour within the 24-hour forecast horizon.

The stacked bars illustrate the relationship between occupied and available parking spaces. The gray segment represents the number of predicted occupied spaces, while the colored segment indicates the predicted number of free spaces. This visual structure allows users to quickly understand how much capacity remains available at each hour.

To improve interpretability, the free parking capacity is categorized into different availability levels based on the ratio between free spaces and the total

CHAPTER 4. RESULTS AND EVALUATION

	Gradient Boosting	Random Forest	Local Weighted Blend Hybrid	Best Model
Forecast Horizon (Hours)				
1	2.42	3.86	2.41	Local Weighted Blend Hybrid
2	4.26	4.73	4.22	Local Weighted Blend Hybrid
3	4.77	4.98	4.69	Local Weighted Blend Hybrid
4	4.65	4.75	4.50	Local Weighted Blend Hybrid
5	4.46	4.63	4.40	Local Weighted Blend Hybrid
6	4.51	4.58	4.34	Local Weighted Blend Hybrid
7	4.54	4.61	4.39	Local Weighted Blend Hybrid
8	4.61	4.64	4.43	Local Weighted Blend Hybrid
9	4.80	4.65	4.53	Local Weighted Blend Hybrid
10	4.94	4.75	4.67	Local Weighted Blend Hybrid
11	4.95	4.93	4.81	Local Weighted Blend Hybrid
12	5.05	5.11	4.98	Local Weighted Blend Hybrid
13	5.33	5.28	5.19	Local Weighted Blend Hybrid
14	5.42	5.47	5.33	Local Weighted Blend Hybrid
15	5.73	5.60	5.53	Local Weighted Blend Hybrid
16	5.83	5.71	5.63	Local Weighted Blend Hybrid
17	5.89	5.78	5.68	Local Weighted Blend Hybrid
18	5.90	5.80	5.73	Local Weighted Blend Hybrid
19	5.86	5.81	5.74	Local Weighted Blend Hybrid
20	5.85	5.83	5.76	Local Weighted Blend Hybrid
21	5.90	5.82	5.74	Local Weighted Blend Hybrid
22	6.02	5.83	5.77	Local Weighted Blend Hybrid
23	5.98	5.84	5.75	Local Weighted Blend Hybrid
24	5.96	5.92	5.76	Local Weighted Blend Hybrid

Figure 4.6: Horizon-wise best model for Lot 634.

parking capacity. The color coding represents the likelihood of finding a free parking space at a given time:

- **Green (Very High Availability):** more than 50% of spaces are free
- **Yellow (High Availability):** between 30% and 50% of spaces are free
- **Orange (Low Availability):** between 15% and 30% of spaces are free
- **Red (Very Low Availability):** less than 15% of spaces are free

This classification enables users to quickly assess the probability of finding a parking space at different times of the day without requiring detailed numerical interpretation. By presenting the predicted occupancy levels together with the categorized availability indicators, the visualization provides a clear overview of expected parking conditions for both parking facilities over the upcoming 24-hour period.

4.4.2 Historical and Predicted Parking Availability Trends

In addition to the categorical availability visualization, the dashboard provides a continuous trend chart that combines recent historical observations with future predictions of parking availability (see Figure 4.11). For each parking facility, the chart displays the number of available parking spaces over a 72-hour window consisting of the previous 48 hours of historical data and the next 24 hours of predicted values.

The historical portion of the chart is shown using a solid line and represents observed parking availability retrieved from the parking data collection service. The predicted portion is displayed using a dashed line with a different color

4.4. DEPLOYMENT RESULTS: INTERACTIVE FORECAST DASHBOARD

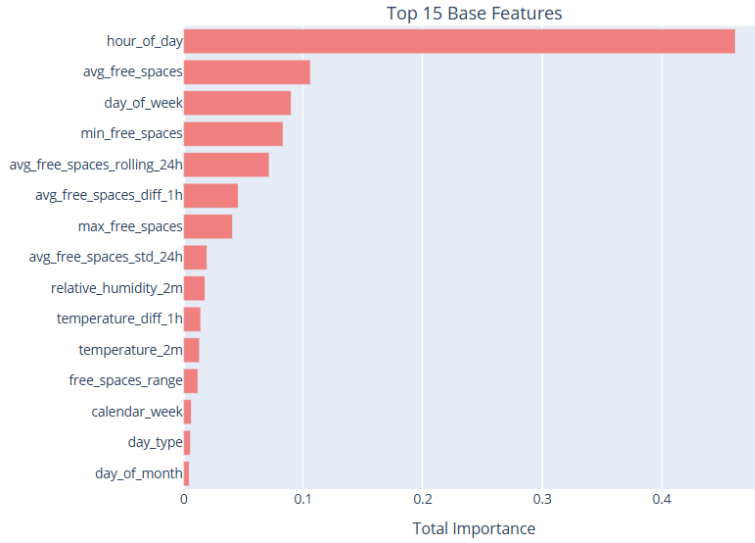


Figure 4.7: Top 15 important features for Lot 38 (Random Forest).

to clearly distinguish forecasted values from observed data. A vertical separator marks the current time, indicating the transition point between historical observations and model predictions.

By presenting both historical and predicted values within a single visualization, the chart allows users to understand recent parking patterns and how these patterns are expected to evolve over the next 24 hours. This continuous representation helps users identify temporal trends, such as periods of increasing or decreasing parking availability, while also providing context for the predicted values based on recent system behavior.

The trend visualization is provided for both monitored parking facilities, allowing users to compare the recent occupancy dynamics and expected availability across locations.

4.4.3 24-Hour Temperature Forecast Visualization

In addition to parking availability predictions, the dashboard also presents the expected temperature trend for the next 24 hours in Bamberg. This information provides contextual environmental conditions that may influence parking demand and mobility patterns.

The temperature forecast is obtained through an external weather API, which is the same service used for collecting meteorological input data in the forecasting pipeline. Instead of generating temperature predictions within the system, the dashboard retrieves the forecasted values directly from the API and displays them to users.

The visualization is presented as a continuous line chart showing the expected temperature values for the upcoming 24-hour period. Each point on the curve represents the forecasted temperature for a specific hour. By presenting this information alongside parking availability predictions, users can interpret expected parking conditions together with the anticipated weather situation.

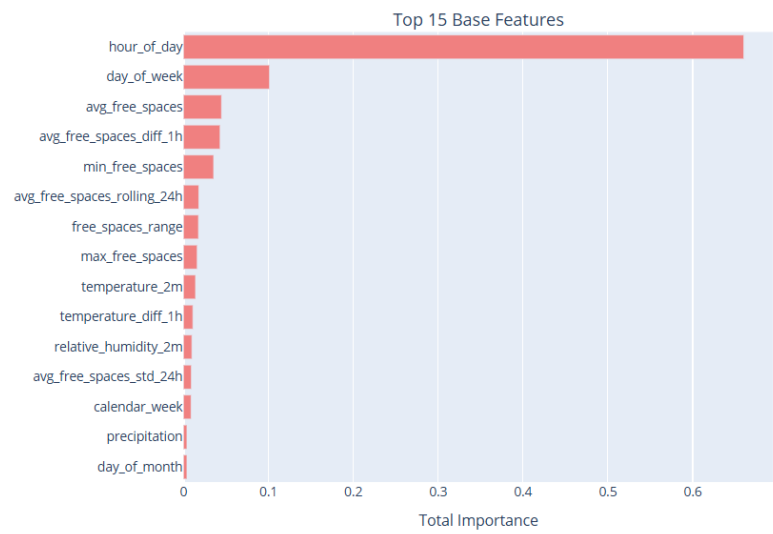


Figure 4.8: Top 15 important features for Lot 634 (Random Forest).

Figure 4.12 illustrates the temperature forecast visualization integrated into the dashboard interface.

4.4. DEPLOYMENT RESULTS: INTERACTIVE FORECAST DASHBOARD



Figure 4.9: Web dashboard for real-time parking availability and 24-hour forecasts.



Figure 4.10: Predicted parking availability for Lots 38 and 634.



Figure 4.11: Historical (48h) and predicted (24h) parking availability for Lots 38 and 634.

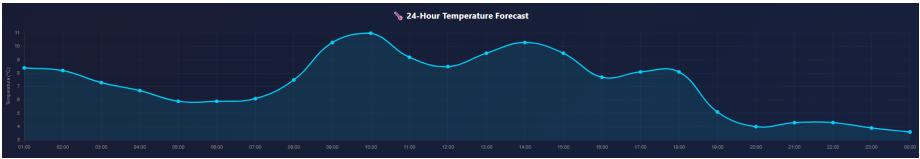


Figure 4.12: 24-hour temperature forecast for Bamberg.

Chapter 5

Discussion and Conclusion

5.1 Discussion

This section interprets the empirical results presented in the previous chapter and discusses their implications in relation to the objectives of this thesis. While the results chapter focused on the quantitative evaluation of different forecasting models, the discussion aims to place these findings into a broader context. In particular, the observed performance patterns across prediction horizons are examined, and the effectiveness of the proposed horizon-aware hybrid forecasting approach is analyzed. Furthermore, the results are discussed in relation to the research questions formulated in the introduction, highlighting both theoretical insights and practical implications for data-driven parking management and decision-support systems. By reflecting on the modeling outcomes and the implemented system architecture, this section provides a deeper understanding of the strengths and limitations of the proposed approach within the domain of urban parking prediction.

5.1.1 Interpretation of the Forecasting Results

The empirical evaluation presented in the previous section demonstrates that forecasting performance varies considerably across model types and prediction horizons. This observation highlights an important characteristic of parking availability prediction: the temporal dynamics of parking demand cannot be captured equally well by a single modeling approach across all forecasting horizons. While some models perform strongly in short-term predictions, others are better suited for capturing longer-term temporal patterns.

For short prediction horizons, the results indicate that classical machine learning models often achieve competitive or even superior performance compared to more complex deep learning approaches. This can be explained by the relatively stable short-term dynamics of parking availability, where recent observations and local temporal features provide strong predictive signals. Tree-based models and linear methods are particularly effective in this context because they can efficiently exploit structured input features derived from recent time windows without requiring large amounts of training data.

As the prediction horizon increases, however, the forecasting task becomes more challenging due to increasing uncertainty and the influence of broader

temporal patterns. In these cases, models capable of capturing longer temporal dependencies become more advantageous. Deep learning approaches, such as recurrent neural networks, are designed to model sequential dependencies and therefore show improved performance for longer forecast horizons where simple autoregressive relationships are insufficient.

A central finding of this thesis is the effectiveness of the proposed horizon-aware hybrid forecasting approach. The results show that the Local Weighted Blend model consistently achieves lower prediction errors than the individual baseline models across both evaluated parking facilities. This improvement can be attributed to the complementary strengths of the underlying models. Instead of relying on a single forecasting technique, the hybrid model dynamically combines predictions from different models depending on their relative performance across horizons. By leveraging the strengths of multiple modeling paradigms, the hybrid approach reduces the risk of systematic errors that may occur when relying solely on a single model architecture.

Another relevant observation is the variation in model performance across the two analyzed parking facilities. Although the overall ranking of models remains relatively consistent, the magnitude of forecasting errors differs between locations. This suggests that local characteristics of parking facilities, such as demand variability, capacity constraints, and temporal usage patterns, influence the predictability of parking availability. Parking facilities with more regular usage patterns are generally easier to forecast, whereas highly dynamic demand patterns increase prediction uncertainty.

Furthermore, the composition of the top-performing baseline models differs between the two parking lots. The four strongest baseline models identified for each facility are not identical, indicating that different modeling approaches capture the dynamics of each parking environment more effectively. Interestingly, while the proposed Local Weighted Blend approach achieves the best predictive performance for both parking facilities, it does so by combining different underlying models in each case. This further highlights the importance of adapting forecasting strategies to location-specific characteristics rather than relying on a single universal model architecture.

Overall, the results confirm that parking availability forecasting is a complex time-series prediction task influenced by multiple temporal and contextual factors. The findings suggest that hybrid forecasting strategies represent a promising approach for addressing these challenges, as they combine the strengths of different model families and allow the forecasting system to adapt to varying prediction horizons and data characteristics.

The feature importance analysis based on the Random Forest model provides valuable insights into the key factors driving parking availability prediction for both parking facilities. A dominant observation across both Lot 38 and Lot 634 is the overwhelming importance of the *hour_of_day* feature. This indicates that parking availability is strongly governed by daily temporal patterns, reflecting recurring demand cycles such as commuting hours, business activities, and nighttime periods. The significantly higher importance score of this feature compared to all others highlights the highly structured and periodic nature of parking behavior. In addition to intra-day patterns, the *day_of_week* feature also plays a crucial role, particularly for Lot 634. This suggests that weekly usage patterns, such as differences between weekdays and weekends, further influence parking demand. The stronger contribution of this

feature in Lot 634 indicates that this parking facility exhibits more pronounced weekly variability compared to Lot 38. Historical availability features, including *avg_free_spaces*, *min_free_spaces*, and short-term lagged differences such as *avg_free_spaces_diff_1h*, are also among the most important predictors. These features capture recent system states and short-term trends, confirming that parking availability has a strong autoregressive component. In particular, the importance of rolling statistics such as *avg_free_spaces_rolling_24h* indicates that incorporating temporal smoothing helps capture underlying demand trends beyond immediate fluctuations. In contrast, weather-related features such as temperature and relative humidity exhibit relatively low importance in both parking facilities. This suggests that, within the analyzed dataset, parking behavior is primarily driven by temporal and historical factors rather than external environmental conditions. Similarly, calendar-based features such as *day_of_month* and *calendar_week* contribute minimally, indicating that fine-grained monthly patterns have limited predictive value. Another notable observation is the difference in feature importance distributions between the two parking facilities. While the overall ranking of key features remains consistent, Lot 634 shows a stronger reliance on weekly patterns, whereas Lot 38 places slightly more emphasis on aggregated availability statistics. This difference reflects the influence of location-specific characteristics, such as demand variability and usage regularity, on the predictive structure of the data.

Overall, the feature importance analysis confirms that parking availability prediction is predominantly driven by temporal regularities and recent system states. These findings highlight the importance of incorporating both time-based features and historical availability information in forecasting models, while also suggesting that external factors such as weather may play a secondary role in this context.

Beyond model performance, an important contribution of this work lies in the integration of the forecasting approach into a real-time dashboard system. While many existing studies focus solely on offline evaluation, the proposed system demonstrates how predictive models can be embedded into a practical decision support tool.

By providing 24-hour forecasts of parking availability, the dashboard enables users to make informed arrival-time decisions rather than relying on reactive, real-time information. This directly addresses the limitations identified in current parking systems, where drivers only have access to present availability and may still experience search time.

The integration of forecasting models into an interactive and continuously updated dashboard highlights the potential of combining machine learning with real-time data pipelines to support urban mobility. This demonstrates that the value of prediction models extends beyond accuracy metrics, contributing to improved usability and real-world applicability.

5.1.2 Answering the Research Questions

This subsection revisits the research questions formulated in the introduction and discusses how the empirical results obtained in this study address them.

RQ1: Which variables have the greatest impact on parking availability prediction?

The feature importance analysis reveals that temporal variables and recent occupancy statistics are the most influential predictors of parking availability. In particular, the variable *hour of day* emerges as the dominant feature, reflecting strong daily usage cycles in urban parking demand. The variable *day of week* also plays a significant role, indicating systematic differences between weekday and weekend patterns. In addition, features capturing recent parking conditions, such as rolling averages and minimum free space statistics, contribute substantially by modeling short-term dynamics. In contrast, weather-related variables and broader calendar indicators exhibit comparatively lower importance. These findings suggest that parking availability is primarily driven by regular temporal patterns and recent utilization behavior.

RQ2: How does forecasting performance vary across different prediction horizons?

The experimental results show that forecasting performance generally decreases as the prediction horizon increases. Short-term forecasts benefit from strong temporal autocorrelation and recent occupancy information, which provide reliable predictive signals. As the horizon extends, uncertainty increases and the influence of recent observations diminishes. Furthermore, different model families exhibit varying strengths across horizons: classical machine learning models perform competitively in short-term predictions, while models capable of capturing sequential dependencies are more effective for longer horizons.

RQ3: Can a horizon-aware hybrid forecasting model improve predictive accuracy compared to single-model approaches?

The evaluation confirms that the proposed horizon-aware hybrid forecasting approach improves prediction accuracy compared to individual baseline models. In particular, the Local Weighted Blend (LWB) consistently achieves lower error values across parking facilities. This improvement is attributed to the model's ability to combine complementary strengths of different forecasting approaches and adapt to horizon-specific performance characteristics, rather than relying on a single model across all prediction steps.

RQ4: How can a parking availability forecasting framework be integrated into a real-time data pipeline that ensures continuous operation and reproducibility?

The results demonstrate that the proposed framework can be successfully integrated into a fully automated real-time pipeline. The system performs continuous data ingestion from parking sensors and weather APIs, processes the data within a prediction pipeline, and generates forecasts deployed on a cloud-based infrastructure. The forecasting service is hosted on an EC2 instance and orchestrated using containerized components, ensuring reproducibility and scalability. This end-to-end implementation highlights the feasibility of deploying forecasting models in operational environments beyond offline experimentation.

RQ5: How can a predictive parking dashboard support decision-making for drivers, city planners, and parking operators?

The developed dashboard provides actionable insights by visualizing both historical and predicted parking availability. Drivers can use short-term forecasts to estimate the likelihood of finding free parking spaces and adjust their behavior accordingly, potentially reducing search traffic and congestion. At a broader level, city planners and parking operators can leverage these insights

to monitor demand patterns, optimize parking management strategies, and improve overall urban mobility.

5.1.3 Theoretical Implications

Beyond the empirical evaluation of forecasting models, the findings of this study contribute several theoretical insights to the literature on parking availability prediction and time-series forecasting in smart mobility systems.

First, the results highlight the importance of horizon-dependent model performance in parking forecasting tasks. The experiments demonstrate that no single model architecture consistently outperforms others across all prediction horizons. Instead, different model types exhibit varying strengths depending on the forecasting horizon and the characteristics of the input features. This observation supports the growing body of literature suggesting that forecasting problems with multi-step prediction horizons benefit from model strategies that explicitly account for horizon-dependent dynamics.

Second, the findings provide empirical evidence for the effectiveness of hybrid forecasting approaches in the context of urban parking prediction. The proposed horizon-aware hybrid model consistently outperforms individual baseline models across the evaluated parking facilities. By combining predictions from multiple models, the hybrid approach leverages complementary strengths of different learning paradigms and reduces the limitations associated with relying on a single model architecture. This supports the theoretical argument that ensemble and hybrid forecasting strategies can improve predictive robustness in complex time-series environments.

Third, the feature importance analysis contributes to the understanding of the underlying drivers of parking availability dynamics. The results indicate that temporal variables and recent occupancy statistics are the most influential predictors, whereas weather variables and broader calendar indicators play a comparatively smaller role. These findings suggest that parking demand is primarily governed by regular temporal mobility patterns and short-term utilization dynamics rather than external contextual variables.

Finally, this work contributes to the emerging research area at the intersection of machine learning and smart mobility systems. While many existing studies focus primarily on model development and offline prediction accuracy, this thesis demonstrates how forecasting models can be embedded within a complete data-driven architecture that integrates data ingestion, machine learning prediction, and interactive visualization. This integration highlights the potential of combining machine learning techniques with real-time data pipelines and decision-support dashboards to enable practical intelligent transportation solutions.

5.1.4 Practical Implications

Beyond its methodological contributions, the proposed forecasting framework also has several practical implications for urban mobility management and intelligent parking systems. By providing reliable short- and medium-term forecasts of parking availability, the system can support multiple stakeholders including drivers, city planners, and smart city platforms.

For drivers, predictive information about future parking availability can significantly improve trip planning. Instead of searching for parking spaces without prior knowledge, drivers can consult the forecasting dashboard to estimate the likelihood of finding free spaces at specific parking facilities. Based on these predictions, drivers can plan their arrival time more effectively or even decide whether visiting a particular parking facility is worthwhile at a given time. This reduces uncertainty during trip planning and helps avoid unnecessary search traffic. As discussed in the introduction, cruising for parking is a major contributor to urban congestion. Providing predictive parking information can therefore reduce traffic volume, decrease fuel consumption, and contribute to lower levels of urban pollution.

For city planners and parking operators, predictive parking analytics offer valuable insights into demand patterns and utilization trends. Forecasting models can help identify peak usage periods, detect recurring demand cycles, and monitor occupancy dynamics across different facilities. Such information can support more informed parking management strategies, including dynamic pricing policies, improved allocation of parking capacity, and better planning of future infrastructure investments. In addition, predictive analytics can assist municipalities in evaluating the effectiveness of parking regulations and demand management policies.

Furthermore, the system architecture developed in this thesis demonstrates how machine learning models can be integrated into broader smart city platforms. The implemented pipeline combines IoT-based parking sensor data, external contextual information such as weather data, automated data processing, and machine learning forecasting within a cloud-based infrastructure. The integration of these components with an interactive dashboard illustrates how predictive analytics can be operationalized in real-world decision-support systems. Such architectures provide a scalable foundation for intelligent mobility services and can be extended to support additional applications such as traffic prediction, mobility demand analysis, or integrated urban transportation management.

The significance of this thesis lies in its contribution to the growing body of research on data-driven parking management by explicitly linking predictive modeling with operational decision support. While parking availability prediction has been widely studied as a forecasting problem, this work reframes it as a decision-support task, where the value of predictions is determined not only by statistical accuracy but also by their usability within real-world systems. By focusing on short-term, 24-hour parking availability forecasts and embedding them into a real-time pipeline and dashboard, the thesis addresses a practical need faced by drivers, city planners, and mobility service providers. This system-oriented perspective contributes to the field by demonstrating how predictive analytics can be transformed into actionable insights that directly support arrival-time planning and infrastructure utilization, rather than remaining confined to offline evaluation.

From an academic standpoint, this thesis contributes to methodological discussions on multi-horizon time series forecasting in intelligent transportation systems. The explicit treatment of horizon-dependent model performance challenges the implicit assumption, still prevalent in much of the literature, that a single model is sufficient for all forecast steps. By systematically analyzing how different models perform across short-term horizons and introducing a horizon-

aware hybrid approach, the thesis provides empirical evidence that supports more nuanced model selection and combination strategies. This contributes to a clearer understanding of when and why certain modeling approaches are effective, enriching comparative forecasting studies with a horizon-sensitive perspective that is often underrepresented in parking prediction research.

Finally, the practical relevance of this thesis provides a strong justification for its contribution to industry and smart city initiatives. The proposed framework demonstrates how sensor-based data, machine learning models, and visualization tools can be integrated into a scalable pipeline that supports continuous operation in a real urban setting. This makes the results transferable to similar parking facilities and urban contexts beyond the Bamberg case study. By aligning predictive modeling with real-time data engineering and user-oriented dashboard design, the thesis offers a blueprint for practitioners seeking to operationalize parking forecasts as part of digital mobility services. As such, the work not only advances academic understanding but also provides tangible value for the design and implementation of intelligent parking decision-support systems.

5.2 Conclusion and Future Work

Urban parking management has become an increasingly complex challenge for modern cities. Limited parking capacity, growing vehicle usage, and inefficient search behavior contribute significantly to urban congestion, fuel consumption, and environmental pollution. Drivers often spend considerable time searching for available parking spaces, which increases traffic density and reduces overall transportation efficiency. These challenges highlight the importance of data-driven solutions that can support both drivers and city planners in making better decisions regarding parking usage and mobility management. In this context, predictive analytics and machine learning techniques offer promising opportunities to forecast parking availability and transform raw sensor data into actionable insights.

The primary objective of this thesis was to develop and evaluate a data-driven forecasting framework capable of predicting parking availability using machine learning techniques and to integrate these predictions into a real-time decision-support system. To achieve this goal, several classical and deep learning models were evaluated for multi-step time-series forecasting of parking availability. In addition to evaluating individual models, this work proposed a horizon-aware hybrid forecasting strategy designed to leverage the complementary strengths of different algorithms. The study also incorporated a feature importance analysis in order to better understand which variables contribute most significantly to parking availability prediction. Finally, the developed forecasting models were integrated into a fully automated pipeline that collects data, generates predictions, and visualizes the results in an interactive dashboard deployed on a cloud-based infrastructure.

The experimental evaluation demonstrates several important findings regarding parking availability forecasting. First, the results confirm that forecasting accuracy varies significantly across prediction horizons. Short-term forecasts generally achieve higher accuracy due to strong temporal autocorrelation and the predictive power of recent occupancy patterns. As the prediction horizon increases, forecasting becomes more challenging due to growing uncertainty and

the influence of more complex temporal dynamics. The experiments further show that different model families exhibit varying strengths across forecasting horizons. While classical machine learning models perform competitively for short horizons due to their ability to exploit structured features and recent statistics, models capable of capturing temporal dependencies become increasingly valuable as the prediction horizon expands.

A key contribution of this thesis is the development and evaluation of a horizon-aware hybrid forecasting model. The results demonstrate that the proposed Local Weighted Blend approach consistently achieves improved predictive performance compared to individual baseline models across the analyzed parking facilities. By combining predictions from multiple models, the hybrid approach reduces the limitations associated with relying on a single forecasting method and allows the system to adapt to varying temporal patterns in the data. Importantly, the experiments reveal that although the hybrid approach performs best for both parking facilities, the composition of the underlying models differs between locations. This observation highlights the influence of location-specific parking dynamics and supports the idea that adaptive model combinations can improve forecasting robustness across heterogeneous environments.

Another important insight gained from this work is provided by the feature importance analysis. The results show that temporal variables and recent occupancy statistics are the most influential predictors of parking availability. In particular, the variable representing the hour of the day emerges as the dominant feature, reflecting strong daily parking usage cycles that correspond to typical urban mobility patterns such as commuting hours and business activities. The day of the week also plays an important role, indicating systematic differences between weekday and weekend demand. In addition, features describing recent parking conditions, such as rolling statistics of free spaces, significantly improve prediction performance by capturing short-term utilization dynamics. In contrast, weather-related variables and broader calendar indicators exhibit comparatively lower importance values in the analyzed dataset. These findings suggest that parking availability is primarily driven by temporal demand patterns and recent occupancy behavior rather than external contextual factors.

Beyond the methodological evaluation of forecasting models, this thesis also demonstrates the practical feasibility of integrating machine learning predictions into a real-time system architecture. The developed framework includes automated data ingestion from parking sensors and external weather APIs, pre-processing and feature generation, model-based prediction, and visualization through an interactive dashboard. The system is deployed on a cloud-based infrastructure and continuously generates forecasts that can be accessed through the visualization interface. This implementation illustrates how predictive analytics can move beyond offline experiments and become part of operational decision-support systems for urban mobility management.

The practical implications of such predictive systems are significant. By providing forecasts of future parking availability, the dashboard enables drivers to better plan their trips and estimate the likelihood of finding available spaces at a specific parking facility. Drivers can adjust their arrival time, select alternative parking locations, or even decide whether visiting a particular location is worthwhile based on predicted availability. As discussed in the introduction, a substantial portion of urban traffic congestion is caused by drivers searching for parking spaces. Providing predictive parking information can therefore help

reduce unnecessary search traffic, which in turn contributes to lower congestion levels, reduced fuel consumption, and decreased urban air pollution.

For city planners and parking operators, predictive parking analytics can provide valuable insights into demand patterns and utilization dynamics. Forecasting models can help identify peak demand periods, detect recurring usage cycles, and support more effective parking management strategies. These insights may support dynamic pricing policies, improved capacity planning, and better allocation of parking infrastructure. Furthermore, predictive parking systems can be integrated into broader smart city platforms that combine multiple sources of urban mobility data in order to support more efficient transportation planning.

Although the proposed framework demonstrates promising results, several limitations should be acknowledged. First, the study focuses on data from two parking facilities, which limits the generalizability of the findings to other cities or parking environments with different characteristics. Parking behavior may vary significantly depending on urban structure, demand patterns, and external conditions.

In addition, the forecasting models rely primarily on temporal and occupancy-related features, while other contextual variables such as major events, road closures, or public transportation disruptions were not incorporated into the prediction models. Including additional contextual data sources may further improve prediction accuracy, particularly for longer forecasting horizons.

From a modeling perspective, the proposed hybrid approach combines two complementary base models for each parking facility. While this design demonstrates strong performance, it represents a simplified configuration. Future work could explore the integration of a larger set of models within the hybrid framework, allowing for more flexible and potentially more robust combinations of forecasting strategies across different horizons.

Another important limitation relates to the operational aspects of machine learning systems. The current implementation assumes a static model after deployment and does not include mechanisms for continuous performance monitoring, data drift detection, or automated model retraining. In real-world applications, parking patterns may evolve over time due to seasonal effects, urban changes, or behavioral shifts. Therefore, incorporating monitoring pipelines, performance tracking, and adaptive retraining strategies is essential to ensure long-term reliability and robustness of the system.

Future research could extend this work in several directions. First, the forecasting framework could be expanded to include a larger number of parking facilities across multiple urban areas in order to evaluate the scalability and robustness of the proposed hybrid forecasting approach. Second, additional contextual data sources such as traffic flow information, event schedules, or mobility patterns could be incorporated to better capture external influences on parking demand. Third, advanced deep learning architectures or probabilistic forecasting methods could be explored to generate not only point predictions but also uncertainty estimates, which may provide more informative decision-support tools. Finally, the integration of predictive parking systems with navigation services or urban mobility applications could further enhance the practical impact of such systems by delivering real-time parking guidance directly to drivers.

In conclusion, this thesis demonstrates the potential of combining machine learning forecasting techniques with real-time data pipelines and visualization

tools to support intelligent parking management. The proposed hybrid forecasting framework improves prediction accuracy while the implemented system architecture illustrates how predictive models can be operationalized within a practical decision-support environment. By transforming parking sensor data into actionable forecasts, the presented approach contributes to the development of smarter urban mobility systems and highlights the value of predictive analytics for addressing real-world transportation challenges.

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Appendix A

Qualitative Prediction Comparison

To provide an illustrative comparison of model behavior, Figures A.1 and A.2 present example prediction trajectories over a 24-hour horizon for both parking facilities. The plots compare the actual future parking availability with the predictions generated by the top-performing baseline models and the proposed hybrid approaches.

The results highlight that while individual models capture the general trend of parking availability, deviations occur at different horizons. The hybrid models, particularly the Local Weighted Blend, more closely follow the actual trajectory by combining the strengths of the base models across short-, medium-, and long-term forecasts.

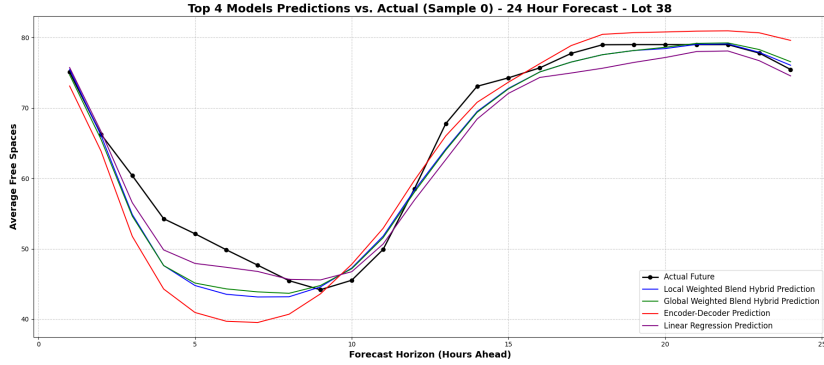


Figure A.1: Example predictions vs. actual values for top models (Lot 38).

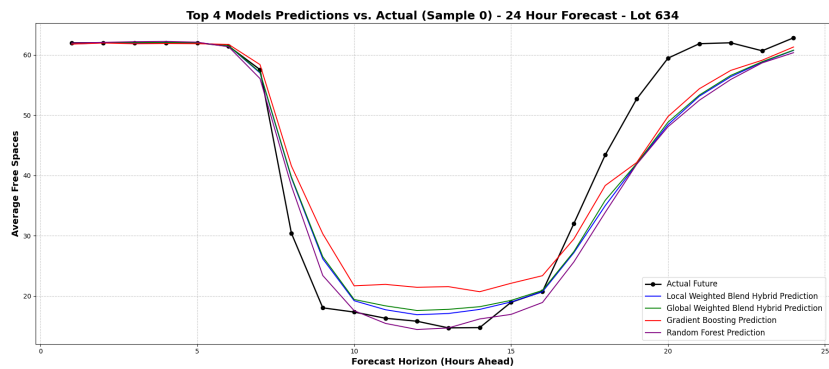


Figure A.2: Example predictions vs. actual values for top models (Lot 634).

Declaration of Authorship

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